

SEASONAL LANDSLIDE INVENTORY FOR NER-2023



North Eastern Regional Node for Disaster Risk Reduction (NER-DRR)
NORTH EASTERN SPACE APPLICATIONS CENTRE
GOVERNMENT OF INDIA, DEPARTMENT OF SPACE
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12	Abstract	The study was taken up as an ongoing process under NERDRR programme of NESAC, Umiam. It is basically a compendium of all reported landslide incidents collected from newspapers and online media reports. A total of 85 landslide incidents have been documented for the different states of the NER for the year 2023, majority of which have been triggered by rainfall. Further, a total of 26 casualties have been reported out of which 4 from Arunachal Pradesh, 1 from Assam, 12 from Meghalaya, 4 Mizoram, 3 Nagaland and 2 Sikkim. In addition, collapse of road-sections of National/State Highways is reported from all the states of NER. This report also contains information on 24 hrs total rainfall prior to each event. However, this report is not inclusive of every landslide events that has occurred during the year 2023. Some of these locations are verified in terms of precise latitude and longitude, either through field survey, photographs and/or post-dated imagery. Accordingly, codification has been assigned to all. The systematic inventory thus compiled will help in carrying out future landslide research. Further this will help in assessing the risk in terms of yearly loss of life and property, disruption of transportation network, frequency of landslide and magnitude of landslide incidents in an area and NER as a whole.		
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PREFACE

North Eastern Regional node for Disaster Risk Reduction (NER-DRR) has been set up at NESAC, Umiam in the year 2011 to provide scientific solutions towards disaster management to the entire region. Almost entire NER consisting of 8 states, namely, Arunachal Pradesh, Assam, Manipur, Meghalaya, Mizoram, Nagaland, Sikkim and Tripura is susceptible to landslides except the plain areas of Assam, Manipur and Tripura. Physiographically NER is apart of Extra-Peninsula of India and of geologically recent origin, having exposed rocks from Archaean to Quaternary ages. This diverse setup makes the terrain susceptible to landslides. While, intense and/or prolonged monsoonal rainfall and seismic activities are the immediate trigger, anthropogenic factors have a role in accelerating the issue. Hence, the need for immediate and systematic cataloguing of landslide incidents is realized by creating a geospatial database highlighting the landslide locations with respect to nearby settlements and transportation network after the event is reported.

It is expected that, the systematic inventory thus compiled will help in carrying out future landslide research. Further, visualization of the same in spatial context will help in assessing the risk in terms of yearly loss of life and property, disruption of transportation network, frequency and magnitude of landslide incidents in an area and NER as a whole.

DPD, NER-DRR

CONTENTS

PREFACE	v
1. INTRODUCTION	1
Indian Scenario	1
Landslides in NER India and its Consequences	2
Consequences of Landslides in North-East India	2
Efforts to Address Landslide Risks	3
2.SOURCES OF DATA	4
Landslide Data	4
Newspapers:	4
Online Media:	5
Satellite Data.....	5
Rainfall Data	7
3. CHARACTERISITICS OF THE LANDSLIDE INVENTORY	8
4. METHODOLOGY	11
Landslide Point and Polygon Extraction	11
Automatic Rainfall Data Extraction	14
5. LANDSLIDE INVENTORY IN DIFFERENT PARTS OF THE NER	14
Arunachal Pradesh.....	14
Assam	16
Manipur	17
Meghalaya	18
Mizoram	20
Nagaland.....	22
Sikkim	23
Tripura	26
6.ANNEXURES	27
Annexure-I:Landslide Incident Reports	27
Annexure-II:Newspaper (Clippings & Digital Archive).....	43
Annexure-III:Online Media.....	45
Annexure-IV:Inventory Attributes	46
Annexure-V:Code Sample for Landslide Detection	47
Annexure-VI:Code Sample for Landslide News	48

LIST OF FIGURES

Figure 1: Sentinel 2 (1C) data overview	4
Figure 2: Sentinel 2 (2A) data overview	4
Figure 3: Band details of Sentinel 2 product	7
Figure 4: GPM IMERG Late Run Product Overview	8
Figure 5: Flowchart for landslide polygon extraction.....	4
Figure 6: Landslides and casualties of NER, India in 2023.....	4
Figure 7: Monthwise landslides in NER, India (2023)	7
Figure 8: Flowchart for automatic extraction of rainfall	8

LIST OF TABLES

Table 1: Sources of landslide data for different states of NER.....	2
Table 2: Statewise list of newspapers reporting landslide incidents in NER	3
Table 3: Landslide statistics of NER, India in 2023	9
Table 4: Landslide casualties of NER, India in 2023	12
Table 5: Summary of landslide incidents in Arunachal Pradesh	14
Table 6: Summary of landslide incidents in Assam.....	16
Table 7: Summary of landslide incidents in Manipur.....	18
Table 8: Summary of landslide incidents in Meghalaya.....	20
Table 9: Summary of landslide incidents in Mizoram.....	14
Table 10: Summary of landslide incidents in Nagaland	16
Table 11: Summary of landslide incidents in Sikkim.....	18
Table 12: Summary of landslide incidents in Tripura	20

LIST OF MAPS

Map 1: Landslide incidents in the NER, 2023.....	6
Map 2: Landslide incidents in Arunachal Pradesh, 2023.....	10
Map 3: Landslide incidents in Assam, 2023	13
Map 4: Landslide incidents in Manipur, 2023	15
Map 5: Landslide incidents in Meghalaya, 2023	17
Map 6: Landslide incidents in Mizoram, 2023	19
Map 7: Landslide incidents in Nagaland, 2023.....	21
Map 8: Landslide incidents in Sikkim, 2023	21
Map 9: Landslide incidents in Tripura, 2023.....	21

1. INTRODUCTION

A 'landslide' is a geological phenomenon characterized by the sudden and rapid movement of a mass of rock, soil, debris or slope forming materials down a slope (Varnes 1958). This movement is driven by the force of gravity, and it can occur in various forms, such as rock slides/falls, debris slides/flows, and earth slides, etc. Landslides can be triggered by natural factors, such as heavy rainfall, snowmelt, earthquakes, volcanic eruptions, and erosion, as well as human activities, including deforestation, construction, mining, and improper land use (Van Westen 2000). The extent of damage caused by landslides can vary from minor disturbances to catastrophic events that lead to significant loss of life and destruction of property and infrastructure. Understanding the underlying causes and mechanisms of landslides is essential for effective hazard assessment, prevention, and mitigation efforts to minimize their impact on vulnerable regions and communities. According to the official data provided by the United Nations International Strategy for Disaster Reduction (UN/ISDR) and the Centre for Research on the Epidemiology of Disasters (CRED) in 2006, landslides ranked as the third most deadly natural disaster (Vordzorgbed 2022). Approximately 4 million individuals were impacted by landslides, as reported by the Office of U.S. Foreign Disaster Assistance (OFDA) and CRED in 2006 (Jain et al. 2023). Landslides account for approximately 17% of all fatalities resulting from natural hazards worldwide (Raja et al. 2017).

Indian Scenario

In the Indian scenario, landslides are a recurrent and significant natural hazard that poses a considerable threat to life, infrastructure, and the environment. The country's diverse topography, coupled with heavy monsoon rains and seismic activity in certain regions, makes India susceptible to landslide occurrences. Landslides pose a significant hazard across approximately 0.42 million sq. km, equivalent to 12.6% of India's land area, excluding snow-covered regions. Among the vulnerable areas, 0.18 million sq. km lie in the North Eastern Himalaya, encompassing Darjeeling and Sikkim Himalaya, while 0.14 million sq. km are found in the North Western Himalaya, including Uttarakhand, Himachal Pradesh, and Jammu & Kashmir. Additionally, 0.09 million sq. km of susceptible land are situated in the Western Ghats and Konkan hills, spanning Tamil Nadu, Kerala, Karnataka, Goa, and Maharashtra (Jain et al. 2023). The remaining 0.01 million sq. km of risk-prone region exists in the Eastern Ghats of Araku in Andhra Pradesh, as reported by the Geological Survey of India (www.gsi.gov.in). Landslides in India have resulted in numerous disasters over the years, causing loss of lives and damage to property and infrastructure. The impact of landslides is not limited to rural areas; urban centres have also witnessed the

consequences of unstable slopes and haphazard development. Efforts to mitigate the risks posed by landslides in India involve various measures, including slope stabilization, early warning systems, and community awareness programs. However, rapid urbanization, deforestation, and improper land-use planning continue to exacerbate the vulnerability of certain regions. Given the potential severity of landslide events, ongoing research, and proactive measures to understand, predict, and manage landslides are crucial for safeguarding lives and promoting sustainable development in landslide-prone areas of India. Notably, the monsoon season is the prime time for landslide occurrences in India, with the Himalayas and Western Ghats being particularly vulnerable due to their hilly terrain and heavy rainfall patterns.

Landslides in NE India and its consequences

Landslides are a prevalent and concerning natural hazard in the North-Eastern region of India. The states of Arunachal Pradesh, Assam, Manipur, Meghalaya, Mizoram, Nagaland, Tripura, and Sikkim are particularly susceptible to landslide occurrences due to their rugged terrain, heavy rainfall, and high seismic activity (Singh 2005). NASA's Landslide Hazard Assessment for Situational Awareness Model (LHASA) classifies Landslide Susceptibility in large parts of North East India under categories 4 and 5 (in a scale from 0 to 5).

Consequences of Landslides in North-East India:

Loss of Lives and Property: Landslides in the region often result in tragic loss of lives, with communities residing in hilly areas being the most vulnerable. These events can also cause extensive damage to houses, infrastructure, and agricultural lands, impacting the livelihoods of local populations.

Disruption of Transport and Communication: Landslides can block roads, highways, and mountain passes, cutting off access to remote regions. This disruption hampers emergency response efforts, hinders the supply chain, and isolates communities.

Environmental Impact: The collapse of slopes during landslides leads to soil erosion and sedimentation in rivers and streams, affecting water quality and aquatic ecosystems. Additionally, deforestation and uncontrolled development contribute to the degradation of natural slopes, exacerbating landslide risks.

Economic Impact: The cost of repairing damaged infrastructure and restoring affected areas after a landslide event can place a significant economic burden on local governments and the country as a whole.

Displacement and Migration: Landslides can force people to abandon their homes, leading to temporary or permanent displacement. This can lead to increased internal migration and overcrowding in nearby urban centres.

Climate Change Amplification: Climate change may intensify rainfall patterns, leading to increased landslide frequency and severity in the future. The region's vulnerability to landslides may further escalate as a consequence of climate change.

Efforts to Address Landslide Risks:

To address the risks associated with landslides in the North-Eastern region of India, several measures have been undertaken:

Hazard Mapping and Risk Assessment: Conducting detailed hazard mapping and risk assessments allows for the identification of high-risk areas and helps land-use planning decisions.

Slope Stabilization and Engineering Solutions: Installing stabilization measures like retaining walls, slope reinforcement, and drainage systems can help mitigate the potential impact of landslides.

Reforestation and Afforestation: Promoting tree planting and forest conservation efforts help prevent soil erosion and stabilize slopes, reducing landslide risks.

Public Awareness and Education: Educating local communities about the dangers of landslides, their causes, and preventive measures can empower them to take proactive steps for their safety.

Addressing the challenges posed by landslides in North-East India requires coordinated efforts from various stakeholders, including government agencies, local communities, and scientific institutions. The North Eastern Regional Node for Disaster Risk Reduction (NER-DRR), NESAC recognizes the urgent need for cataloguing landslide incidents and developing a Geospatial database. This database will help in identifying the locations of landslides concerning nearby settlements and transportation networks, triggered by any reported landslide incident. The comprehensive inventory thus compiled will serve as a valuable resource for future landslide research and will aid in assessing state/area-specific risks, including loss of life and property, disruptions in transportation networks, frequency, and magnitude of landslides across the entire NER region. This concise report presents information on reported landslide incidents gathered from various sources throughout the NER. However, it should be noted that the data may not encompass all events that occurred during the 2023. By prioritizing proactive measures and sustainable land-use practices, the region can work towards minimizing the consequences of landslides and ensuring the safety and resilience of its inhabitants.

2. SOURCES OF DATA

LANDSLIDE DATA

Unlike disasters such as earthquakes, landslide incidents are not currently recorded and reported through instrumental means. As a result, the immediate cataloguing of landslides primarily relies on information sourced from newspapers and online/social media. Subsequently, the reported landslide locations undergo verification for precise latitude and longitude, either through field surveys or by referencing post-dated satellite/UAV imagery whenever available. Each incident is assigned a specific code for identification purposes. The data sources contributing to this inventory are categorized into two main categories:

1. Newspapers and online media
2. Satellite imagery

Particular attention is paid to minimize confusion regarding the occurrence dates and exact incident locations of the reported events.

Table 1: Sources of landslide data for different states of NER

Source	Arunachal Pradesh	Assam	Manipur	Meghalaya	Mizoram	Nagaland	Sikkim	Tripura
Newspapers & Media	19	8	3	22	8	10	14	1

NEWSPAPERS

Newspapers are the primary sources of information that record landslide incidents in NER. Hence, leading daily newspapers distributed in the different states of NER were thoroughly reviewed (Annexure II). In addition, online media plays a significant role in reporting disaster incidents. Therefore, data collected from newspapers were supplemented with information acquired from online media.

State wise list of newspapers that are referred for 2023 landslide incidents in NER are provided in Table 2.

Table 2 Statewise list of newspapers reporting landslide incidents in NER

News Paper and Media	Arunachal Pradesh	Assam	Manipur	Meghalaya	Mizoram	Nagaland	Sikkim	Tripura
Sentinel Assam		✓						
Hindustan Times	✓			✓				
The Times of India	✓	✓		✓	✓		✓	

The Telegraph			✓	✓				
The Arunachal Times	✓							
The Meghalayan		✓		✓				
Nagaland Post						✓		
Highland Post				✓				
Amar Ujala			✓					
Eastern Mirror			✓					
United News of India								✓
The Print					✓			
Barak Outlet								
The Hindustan Gazette							✓	
East Mojo					✓	✓		
India Today	✓	✓		✓			✓	
Northeast Live	✓	✓						
G Plus News					✓			
Zee Business					✓			
Mint	✓				✓			
Zee News							✓	
NDTV							✓	
Arunachal24.in								
ABP Live						✓	✓	
Youtube Clips	✓	✓	✓	✓	✓	✓	✓	✓

ONLINE MEDIA

In the context of this inventory, online media includes all the small local/regional news stations that only have an online presence. With the ubiquitous accessibility of internet, often landslide incidents photographed/recorded by people end up on different social media platforms. Certain community outreach groups also provide information about recent disasters on such sites. The army also provides updates on X (formerly twitter) whenever they help in clearing landslide debris to facilitate movement.

Examples of the same used in this inventory are provided in the Annexure-III.

SATELLITE DATA

Satellite imageries available post event or post monsoons are used to extract landslide areas through polygon features to validate positional accuracy of the landslide locations. Freely available imagery from SENTINEL 2 Multispectral Instrument (MSI) is mainly used for extraction and updation of landslide features. Details of the Level 1C and the Level 2A products of SENTINEL 2 MSI used for this study is given below.

Sentinel-2 Level 1C Top of Atmosphere (TOA) [🔗](#)

[User guide](#) [Catalog API](#)

[resolution](#) 10m [revisit](#) 5-day



[VIEW IN BROWSER](#)

Overview

[Sentinel-2 Level 1C](#) products are available globally from 2015 onwards. These products are resampled with a constant Ground Sampling Distance (GSD) of 10, 20 and 60 m, depending on the native resolution of the different spectral bands. Pixel coordinates refer to the upper left corner of the pixel.

Offered Data

Archive Status	Access Type	Spatial Extent	Temporal Extent	Available in Ecosystem from
Packed or Unpacked	Immediately available data (IAD)	World	Jul 2015 - Present	Jan 2023

Figure 1. Sentinel 2 (1C) data overview

Sentinel-2 Level 2A Top of Canopy (TOC) [🔗](#)

[User guide](#) [Catalog API](#)

[resolution](#) 10m [revisit](#) 5-day

[ARD](#)



[VIEW IN BROWSER](#)

Overview

[Sentinel-2 Level 2A](#) Level 2A product provides atmospherically corrected Surface Reflectance (SR) images, derived from the associated Level-1C products. The atmospheric correction of Sentinel-2 images includes the correction of the scattering of air molecules (Rayleigh scattering), of the absorbing and scattering effects of atmospheric gases, in particular ozone, oxygen and water vapour and the correction of absorption and scattering due to aerosol particles. Level 2A product are considered an ARD product.

Figure 2. Sentinel 2 (2A) data overview

Copernicus Sentinel data 2023 [Cite](#)

Spatial Extent

[-180, -90, 180, 90]

Temporal Interval

['2015-07-04T00:00:00Z', None]

Spectral Bands

Band Name	Common Name	GSD(m)	Center Wavelength(μm)
B01	Coastal aerosol	60	0.443
B02	Blue	10	0.49
B03	Green	10	0.56
B04	Red	10	0.665
B05	Vegetation Red Edge	20	0.705
B06	Vegetation Red Edge	20	0.74
B07	Vegetation Red Edge	20	0.783
B08	NIR	10	0.842
B8A	Vegetation Red Edge	20	0.865
B09	Water vapour	60	0.945
B10	SWIR - Cirrus	60	1.375
B11	SWIR	20	1.61
B12	SWIR	20	2.19
SCL	Scene Classification	20	
SNW	Snow probability	20	
CLD	Cloud probability	20	

Figure 3. Band details of Sentinel 2 product

Both Level 1C and 2A products were used with the Ground Sampling Distance (GSD) of 10m.

RAINFALL DATA

Rainfall data from NASA's Global Precipitation Mission (GPM) has been used, considering the seamless availability of rainfall data for the entire NER. 'IMERG LATE RUN' product of spatial resolution - 0.1 degrees/~10km has been used.

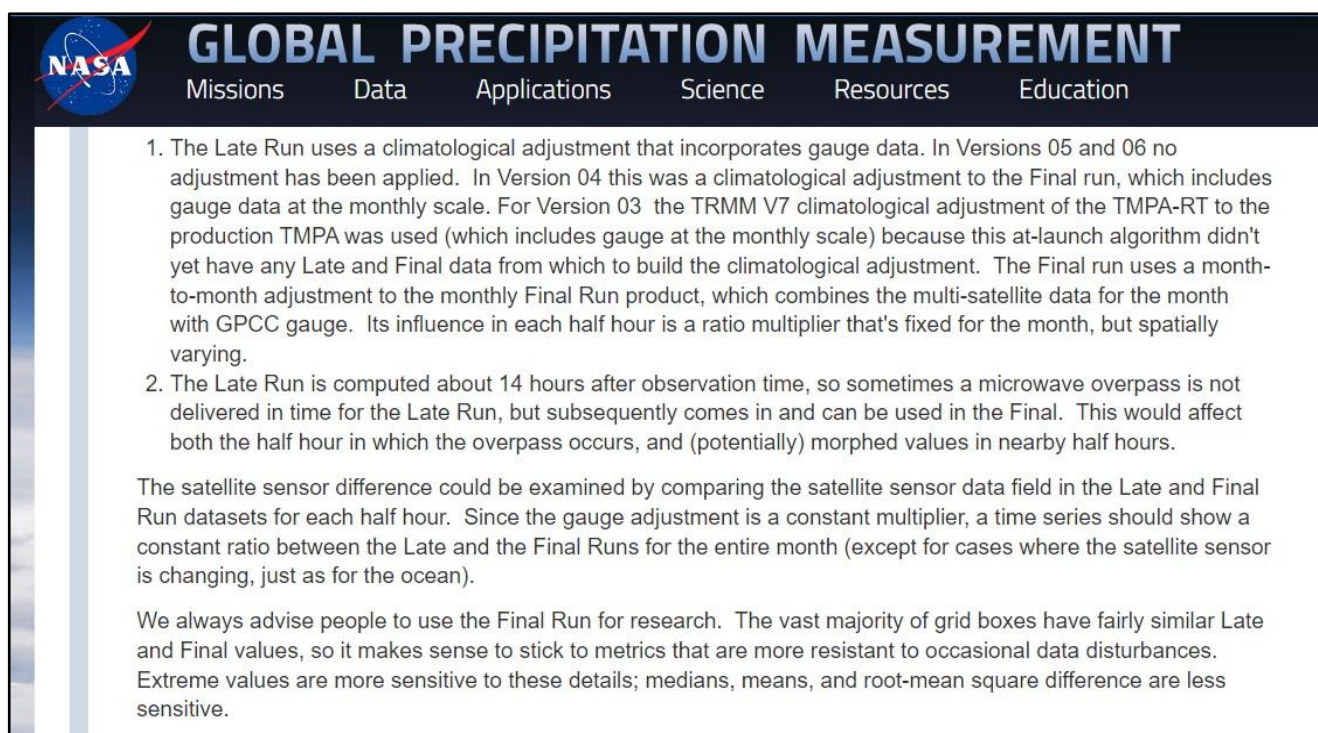


Figure 4. GPM IMERG Late Run Product Overview

3. CHARACTERISTICS OF THE LANDSLIDE INVENTORY

The inventory presented here contains the landslide events that occurred during the year 2023 triggered by monsoonal rainfall. Across the different states of NER India, a total of 85 landslide incidents accounting 26 casualties have been reported. Among these incidents, 19 from Arunachal Pradesh, 8 from Assam, 3 from Manipur, 22 from Meghalaya, 8 from Mizoram, 10 from Nagaland, 14 from Sikkim and 1 from Tripura. Moreover, the inventory also includes reports of road-section collapses along National/State Highways within the NER states. Base and other ancillary information were used to correlate the inventory database is of 1:50,000 scale.

The database captures landslide locations promptly after an event is reported in newspapers, online media, or offline sources, ideally within 24 hours of reporting. Based on the availability of precise location information, the inventory database is categorized into the following two categories given below.

- **TYPE1 (Represented by Green Colour)**

These are the events, for which precise latitude/longitude information is available. Such events are generally validated from UAV/Satellite Imagery and/or Field Visits. Post-dated satellite imageries are used for validation purpose only where spatial resolution of the image, size and exposure condition of the landslide makes it possible.

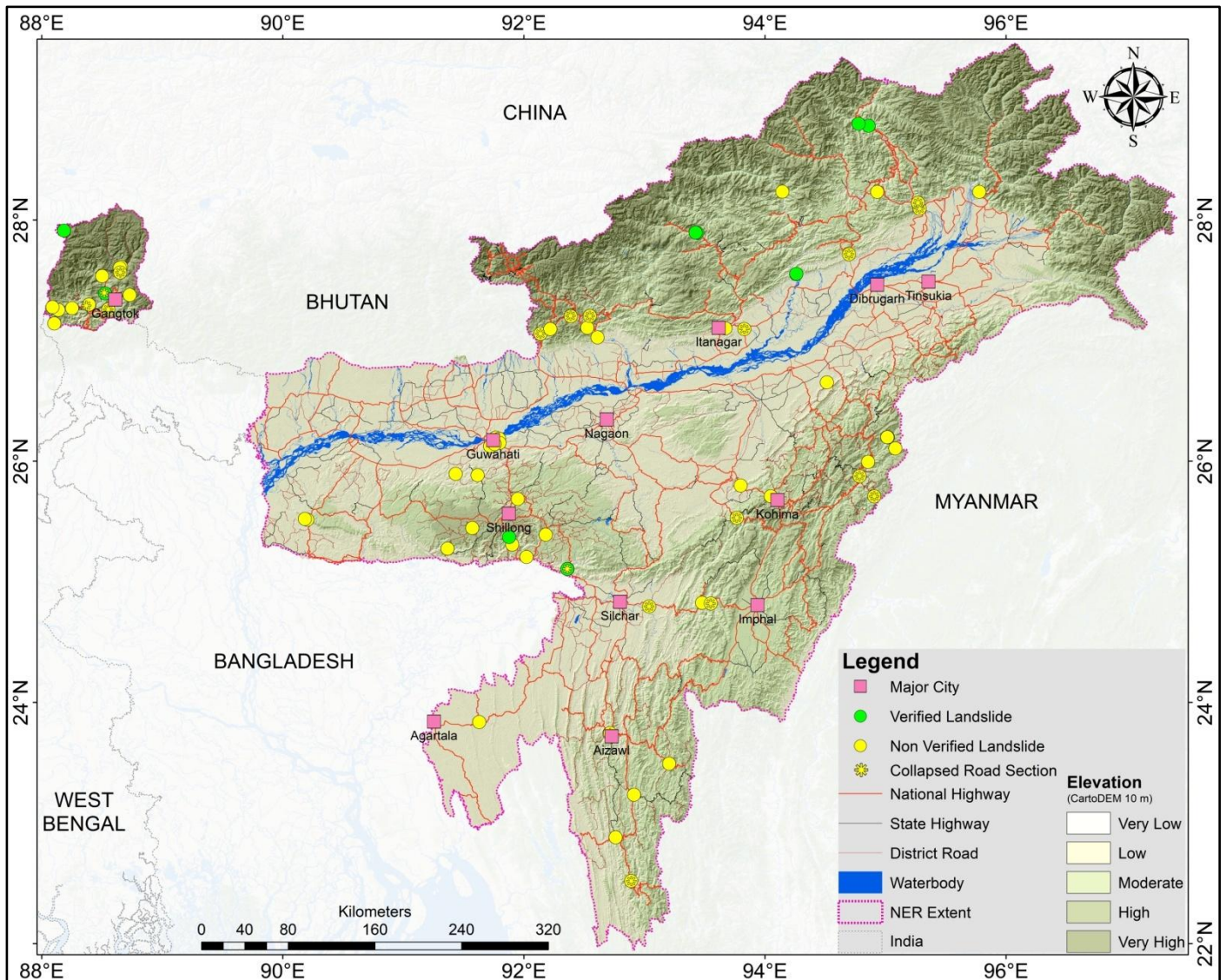
- **TYPE2 (Represented by Yellow colour)**

These are the events, for which precise latitude/longitude information not available rather is referred with respect to nearby village/settlement or locality name by newspapers/media reports. In most of the case these slides are of smaller dimension and could not be visualized using freely available Sentinel 2 data of 10m spatial resolution. However, as and when needed the same may be validated by using post-dated high- resolution satellite data.

- A 3rd category (Type 3) comprises areas where the impact of landslides on road sections along National Highways/State Highways is reported, resulting in transportation disruptions. A distinct symbology is utilized to distinguish these locations. However, validating these areas presents a challenge as road sections are typically restored or repaired promptly, leaving only a limited window for verification through field visits or satellite imagery. Additionally, the revisit status of a particular satellite further complicates the situation. Nevertheless, if required, validation can be conducted using post-dated satellite data with an appropriate resolution.

All the landslide locations also have standard attribute information, in terms of unique ID, location, date/time/type of events, casualty information, and 24 hrs rainfall prior to the event etc. (Annexure IV). Due care is been taken to validate all the event locations at the earliest.

Landslide Locations 2023, NER, India



Map 1: Landslide Incidents in the NER, 2023

4. METHODOLOGY

LANDSLIDE POINT AND POLYGON EXTRACTION

To create the Seasonal Landslide Inventory for the monsoon period in 2023, an extensive review of relevant articles concerning landslide incidents from major Daily Newspapers in NER was conducted. The locations of all landslide incidents were extracted from these newspaper reports, and an initial database was prepared in the form of a Point Shape file. From all the incidents, those deemed significant enough to be visible on Sentinel 2 imagery were isolated, and a 10 km buffer was established around them to resolve any geo-location uncertainties. Subsequently, using Pre and Post Event Satellite Imagery, visible landslide polygons within the 10 km buffer were digitized and added to the inventory.

The flowchart for the methodology is displayed in Figure 4 and the relevant code is provided in the Annexure V.

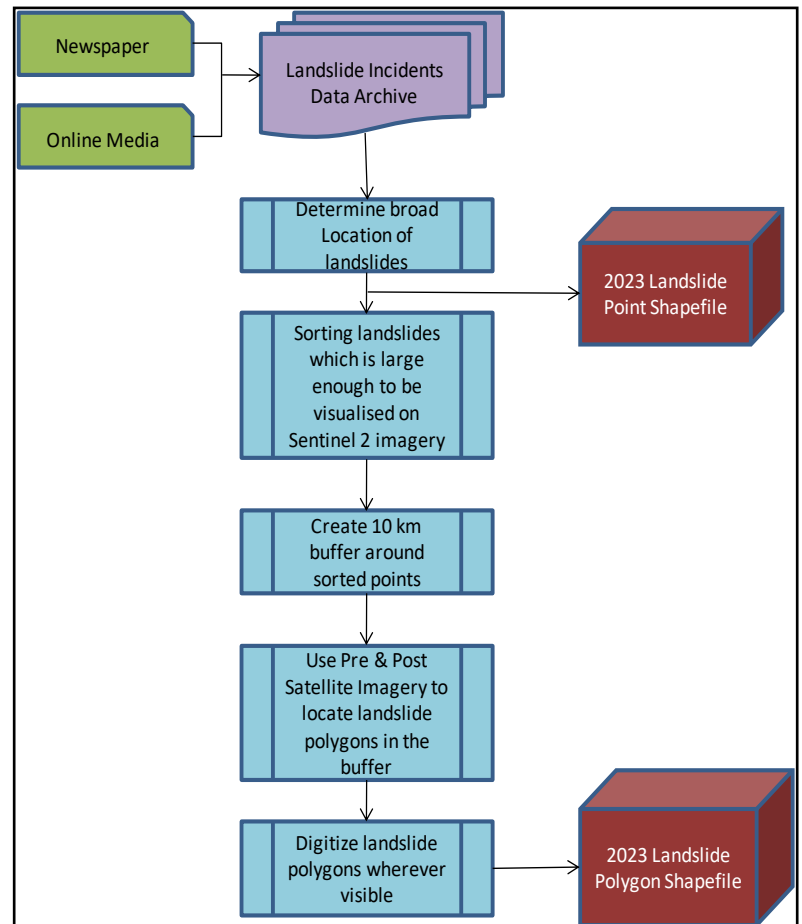


Figure 5. Flowchart for landslide polygon extraction

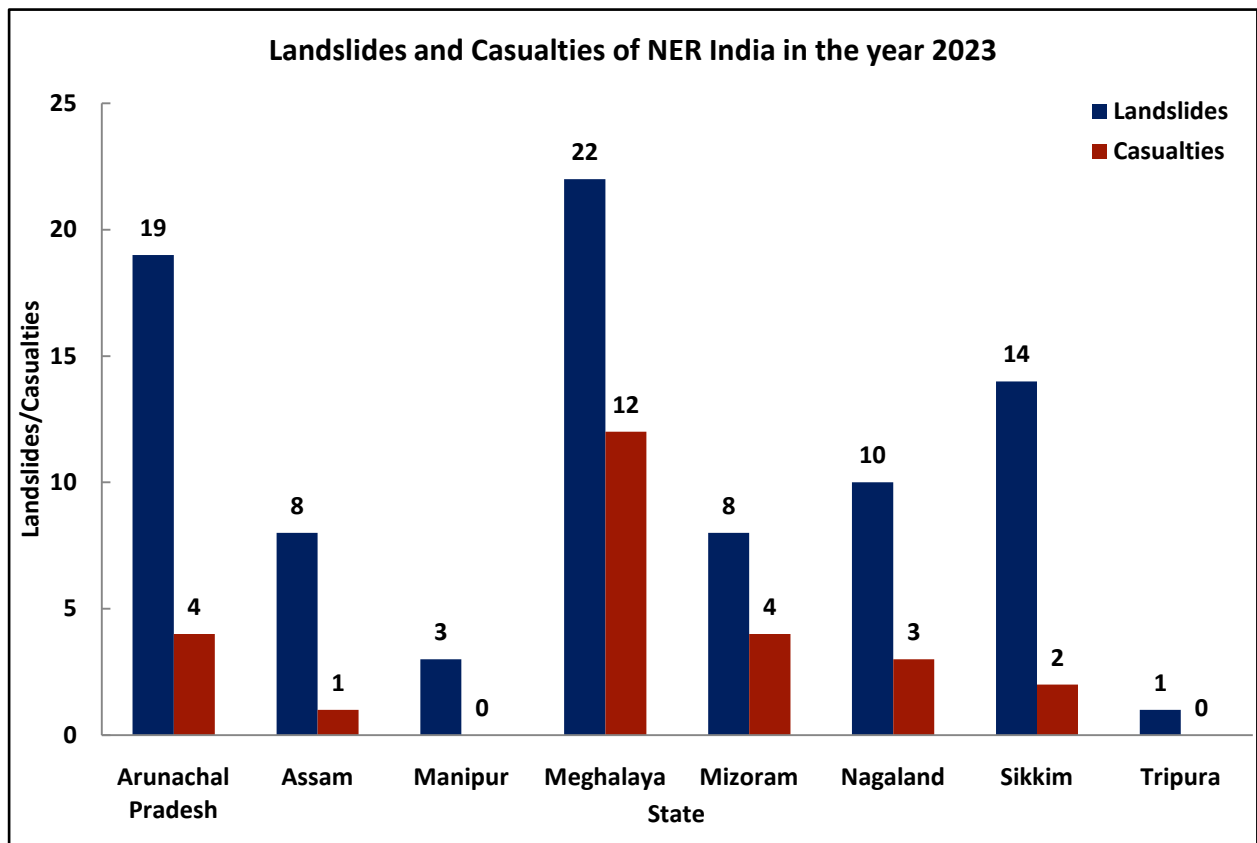


Figure 6. Landslides and casualties of NER, India in 2023

Table 3 Landslide statistics of NER, India in 2023

Months	Arunachal Pradesh	Assam	Manipur	Meghalaya	Mizoram	Nagaland	Sikkim	Tripura	Total Landslides
January	1				1				2
February									0
March	1						1	1	3
April		1		2					3
May	3			2			1		6
June	5	2		12	1		4		24
July	3			1	2	7	4		17
August	5	1	3	3	3	3	3		21
September				1	1				2
October	1	4		1			1		7
November									0
December									0
Total	19	8	3	22	8	10	14	1	85

Table 4 Landslide casualties of NER, India in 2023

Months	Arunachal Pradesh	Assam	Manipur	Meghalaya	Mizoram	Nagaland	Sikkim	Tripura	Total Casualties
January					3				3
February									0
March									0
April				2					2
May	4								4
June		1		6					7
July						3	1		4
August							1		1
September					1				1
October				4					4
November									0
December									0
Total	4	1	0	12	4	3	2	0	26

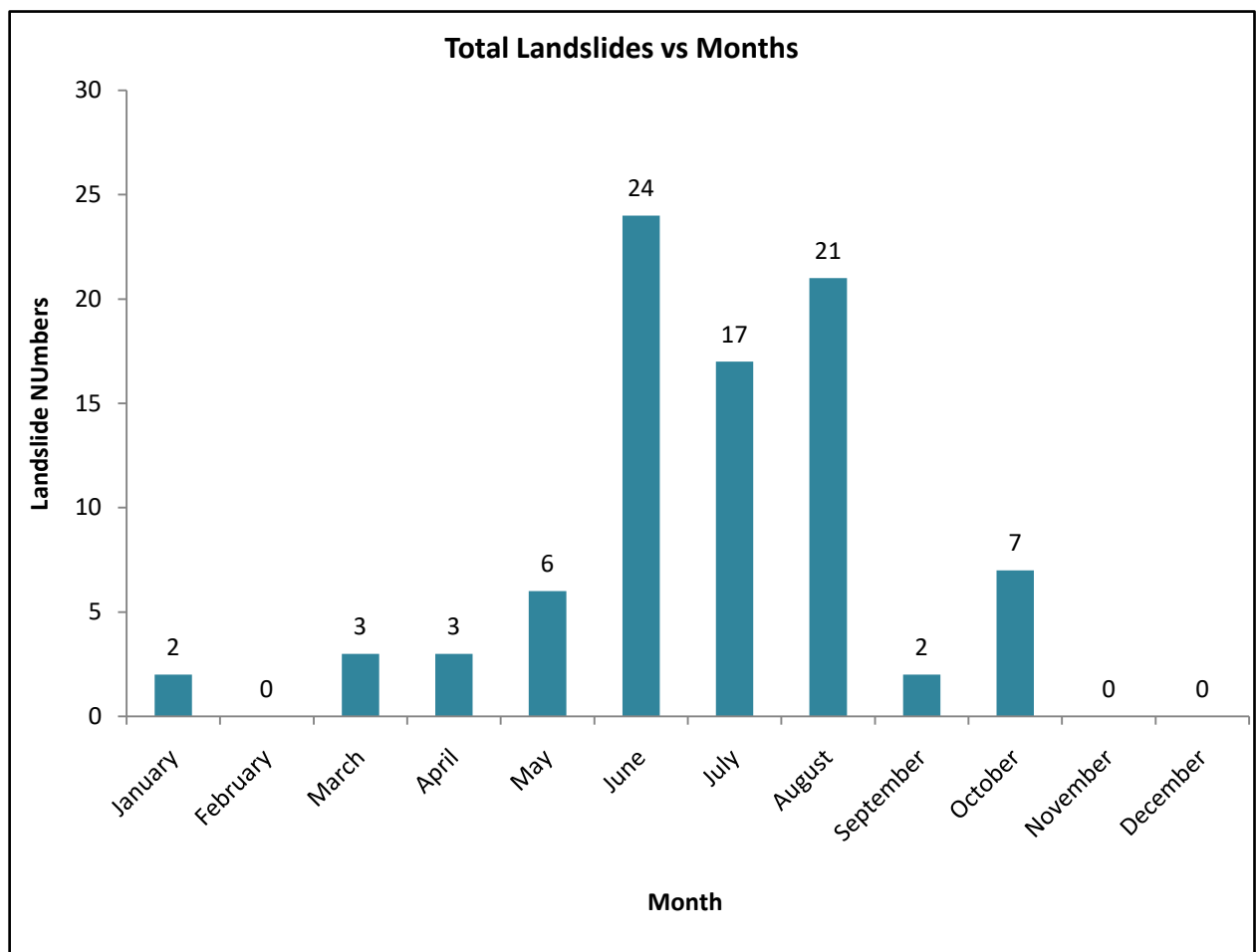


Figure 7. Monthwise landslides in NER, India (2023)

AUTOMATIC RAINFALL DATA EXTRACTION

From the list of landslides events, latitude, longitude and date of event are used to extract the corresponding rainfall data using our created automation pipeline. This automation pipeline will first download the required rainfall data from specific sources (Currently GPM data supported) and pre-process them to get hourly precipitation and cumulative precipitation of each event and finally generate the rainfall intensity graph for each event.

The flowchart for the methodology behind Automatic Rainfall Data Extraction is shown in Figure 5.

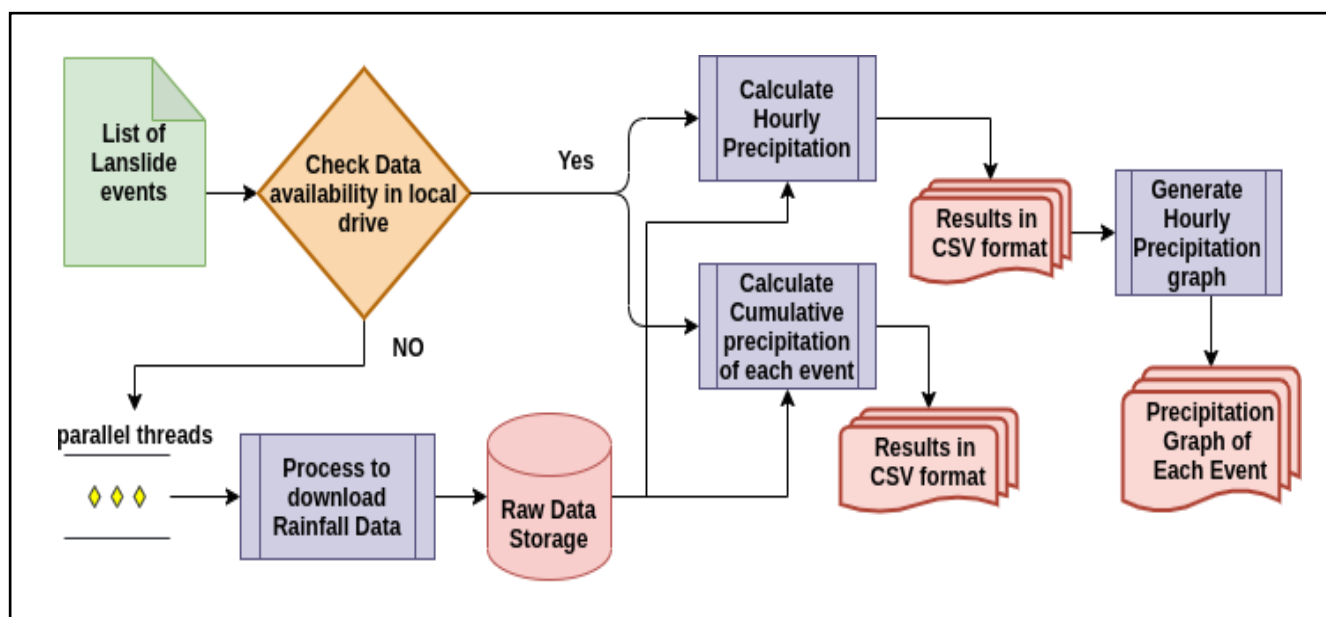


Figure 8. Flowchart for automatic extraction of rainfall

5. LANDSLIDE INVENTORY INDIFFERENT PARTS OF THE NER INDIA

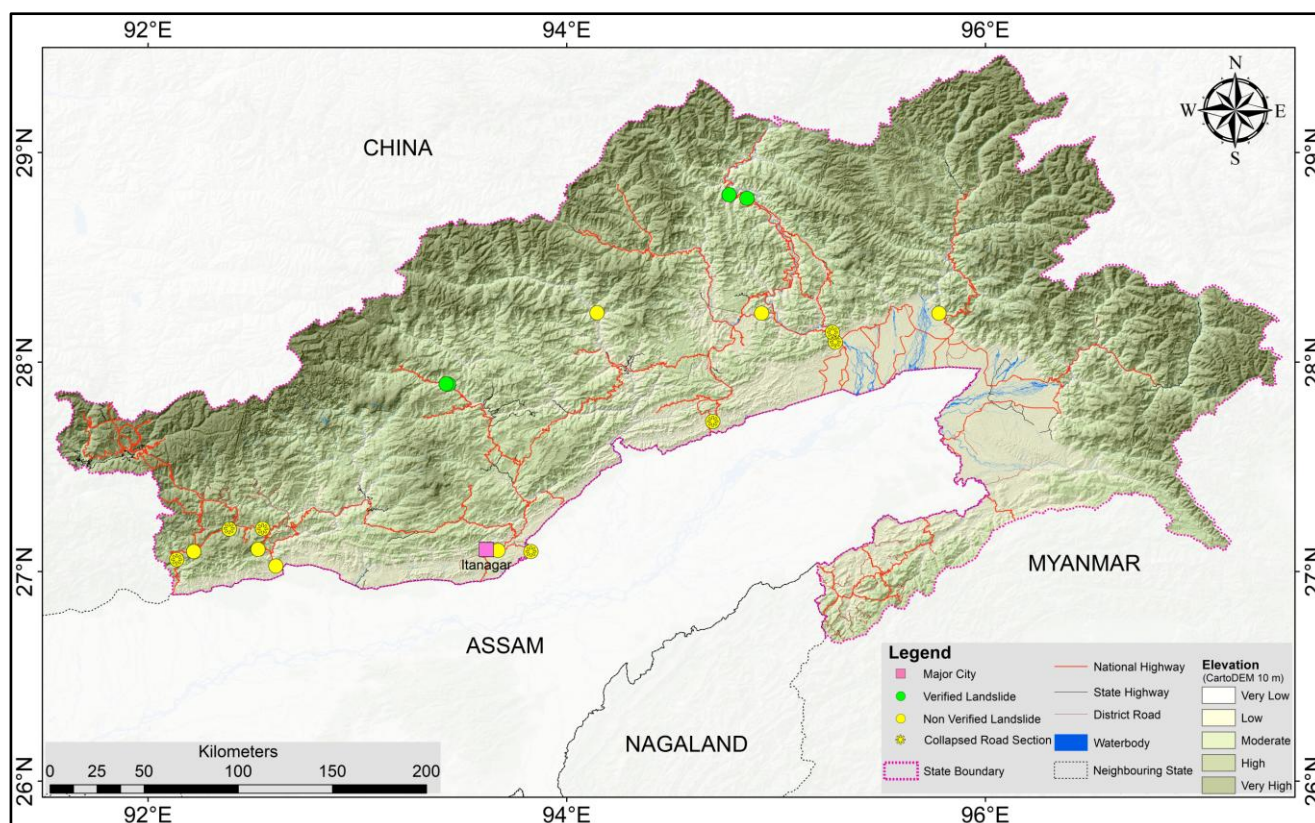
This section of the report depicts a concise overview of landslide incidents across the various states of the NER. Detailed information regarding each landslide incident is provided in the annexure.

ARUNACHAL PRADESH

- A total of 19 landslide incidents recorded in Arunachal Pradesh.
- The first landslide incident for the year was recorded on 24th February 2023 and the latest event was recorded on 27th October 2023.
- A total of 4 casualties occurred in Arunachal Pradesh in 2023.
- NH-415 between Bandardeva and Nirjuli and NH-13 between Pangin and Pasighat were some of the highways/road sections that were affected in the state due to landslides.
- Lower Subansiri Hydroelectric project was partially damaged by landslide.

Table 5 Summary of landslide incidents in Arunachal Pradesh

S. No.	Landslide ID	Source	District	Date	Type	Casualty
1	Newspapers & Media/2023/Arunachal Pradesh/1	Newspapers & Media	Lower Dibang Valley	2023-02-24	2	—
2	Newspapers & Media/2023/Arunachal Pradesh/6	Newspapers & Media	Upper Siang	2023-03-24	3	—
3	Newspapers & Media/2023/Arunachal Pradesh/7	Newspapers & Media	Upper Siang	2023-05-05	3	4
4	Newspapers & Media/2023/Arunachal Pradesh/4	Newspapers & Media	Kurung Kumey	2023-05-06	3	—
5	Newspapers & Media/2023/Arunachal Pradesh/5	Newspapers & Media	Kurung Kumey	2023-05-06	3	—
6	Newspapers & Media/2023/Arunachal Pradesh/17	Newspapers & Media	West Kameng	2023-06-12	2	—
7	Newspapers & Media/2023/Arunachal Pradesh/33	Newspapers & Media	Papum pare	2023-06-23	2	—
8	Newspapers & Media/2023/Arunachal Pradesh/34	Newspapers & Media	Papum pare	2023-06-23	2	—
9	Newspapers & Media/2023/Arunachal Pradesh/35	Newspapers & Media	West Kameng	2023-06-23	2	—
10	Newspapers & Media/2023/Arunachal Pradesh/36	Newspapers & Media	West Siang	2023-06-23	2	—
11	Newspapers & Media/2023/Arunachal Pradesh/50	Newspapers & Media	Siang district	2023-07-16	2	—
12	Newspapers & Media/2023/Arunachal Pradesh/53	Newspapers & Media	Upper Subansiri	2023-07-28	2	—
13	Newspapers & Media/2023/Arunachal Pradesh/54	Newspapers & Media	Siang district	2023-07-28	2	—
14	Newspapers & Media/2023/Arunachal Pradesh/64	Newspapers & Media	Lower Siang	2023-08-18	2	—
15	Newspapers & Media/2023/Arunachal Pradesh/72	Newspapers & Media	West Kameng	2023-08-27	2	—
16	Newspapers & Media/2023/Arunachal Pradesh/73	Newspapers & Media	West Kameng	2023-08-27	2	—
17	Newspapers & Media/2023/Arunachal Pradesh/74	Newspapers & Media	West Kameng	2023-08-27	2	—
18	Newspapers & Media/2023/Arunachal Pradesh/75	Newspapers & Media	West Kameng	2023-08-27	2	—
19	Newspapers & Media/2023/Arunachal Pradesh/85	Newspapers & Media	Lower Subansiri	2023-10-27	2	—



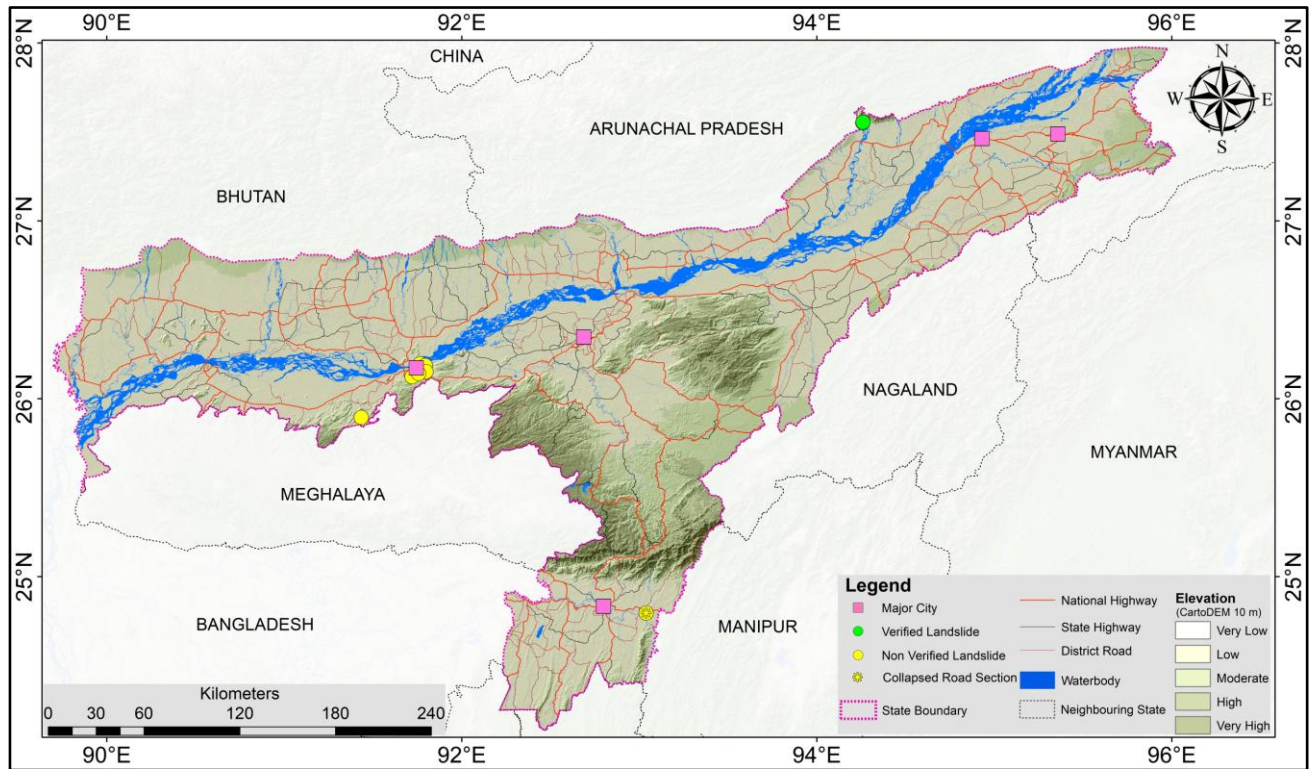
Map 2: Landslide incidents in Arunachal Pradesh, 2023

ASSAM

- A total of 8 landslide incidents recorded in Assam.
- The first landslide incident for the year was recorded on 3rd April 2023 and the latest event was recorded on 7th October 2023.
- A total of 1 casualty occurred in Assam in 2023.
- Landslide in Assam caused lot of damages to properties and injuries to people.

Table 6 Summary of landslide incidents in Assam

S. No.	Landslide ID	Source	District	Date	Type	Casualty
1	Newspapers & Media/2023/Assam/8	Newspapers & Media	Dhemaji	2023-04-03	1	0
2	Newspapers & Media/2023/Assam/23	Newspapers & Media	Kamrup Metro	2023-06-17	2	1
3	Newspapers & Media/2023/Assam/40	Newspapers & Media	Kamrup	2023-06-29	2	0
4	Newspapers & Media/2023/Assam/66	Newspapers & Media	Cachar	2023-08-21	2	0
5	Newspapers & Media/2023/Assam/80	Newspapers & Media	Kamrup Metro	2023-10-06	2	0
6	Newspapers & Media/2023/Assam/81	Newspapers & Media	Kamrup Metro	2023-10-06	2	0
7	Newspapers & Media/2023/Assam/82	Newspapers & Media	Kamrup Metro	2023-10-07	2	0
8	Newspapers & Media/2023/Assam/83	Newspapers & Media	Kamrup Metro	2023-10-07	2	0



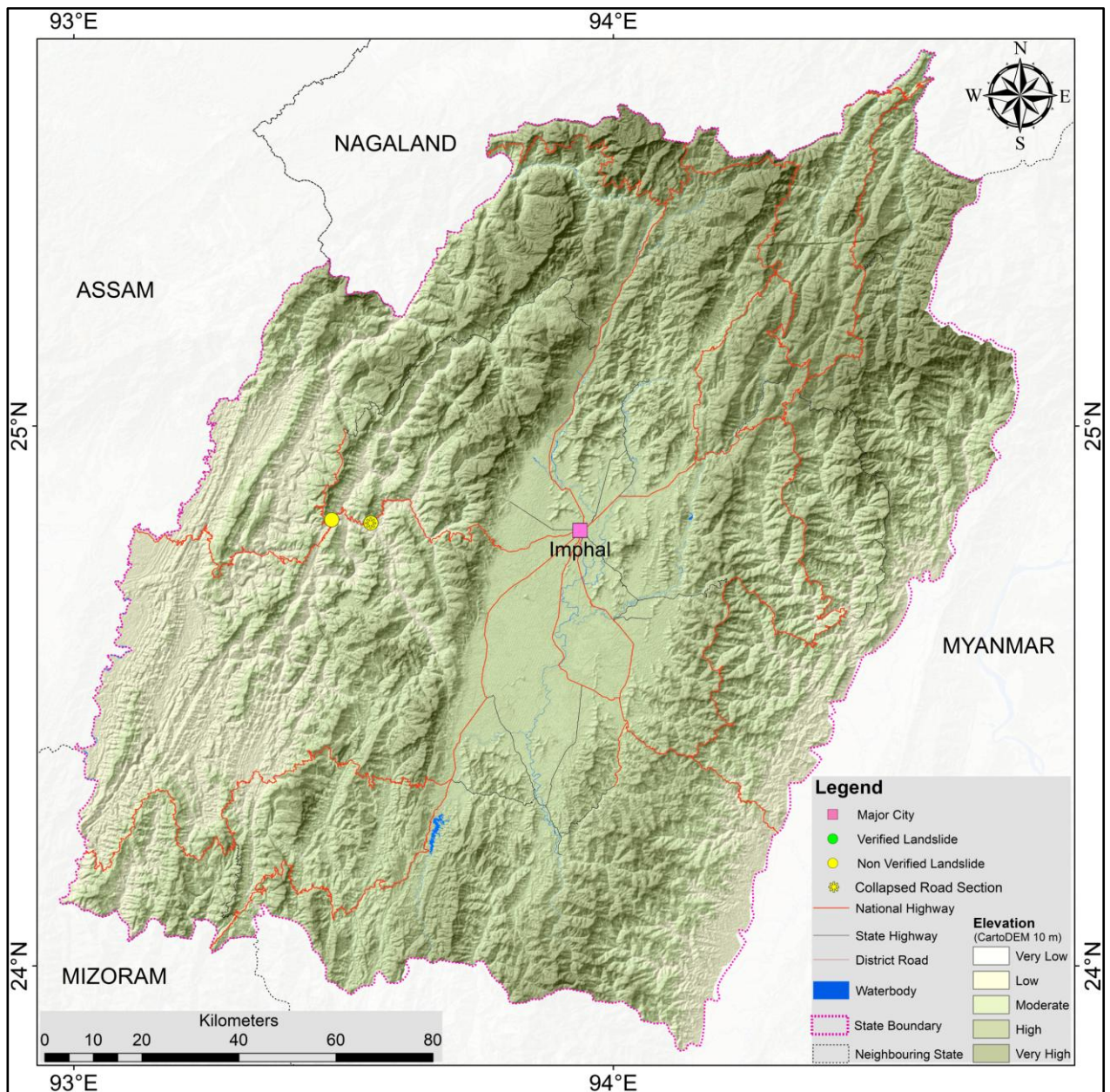
Map 3: Landslide incidents in Assam, 2023

MANIPUR

- A total of 3 landslide incidents recorded in Manipur.
- The first landslide incident for the year was recorded on 16th August 2023 and the latest event was recorded on 20th August 2023.
- Casualties due to landslides were not reported from Manipur in 2023.
- Landslides caused disruptions of traffic in Manipur.

Table 7 Summary of landslide incidents in Manipur

S. No.	Landslide ID	Source	District	Date	Type	Casualty
1	Newspapers & Media/2023/Manipur/62	Newspapers & Media	Noney	2023-08-16	2	0
2	Newspapers & Media/2023/Manipur/63	Newspapers & Media	Noney	2023-08-16	2	0
3	Newspapers & Media/2023/Manipur/65	Newspapers & Media	Noney	2023-08-20	2	0



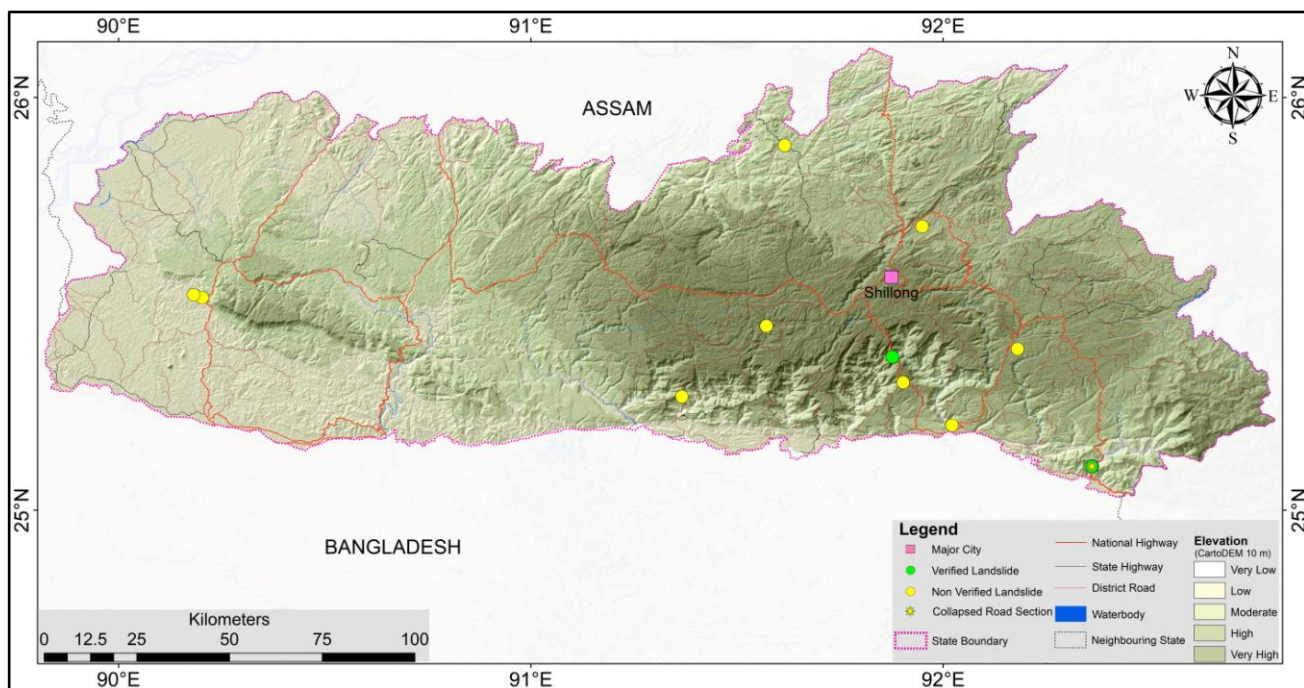
Map 4: Landslide incidents in Manipur, 2023

MEGHALAYA

- A total of 22 landslide incidents recorded in Meghalaya.
- Out of which 10 landslide incidents occurred in Sonapur Tunnel between April to August 2023.
- The first landslide incident for the year was recorded on 5th April 2023 and the latest event was recorded on 8th October 2023.
- 12 casualties were reported from Meghalaya in 2023.
- NH-6 and NH-206 were some of the highways/road sections that were affected in the state due to landslides

Table 8 Summary of landslide incidents in Meghalaya

S. No.	Landslide ID	Source	District	Date	Type	Casualty
1	Newspapers & Media/2023/Meghalaya/14	Newspapers & Media	East Jaintia Hills	2023-04-05	3	0
2	Newspapers & Media/2023/Meghalaya/3	Newspapers & Media	East Khasi Hills	2023-04-14	3	2
3	Newspapers & Media/2023/Meghalaya/9	Newspapers & Media	West Garo Hills	2023-05-06	2	0
4	Newspapers & Media/2023/Meghalaya/12	Newspapers & Media	East Jaintia Hills	2023-05-27	2	0
5	Newspapers & Media/2023/Meghalaya/15	Newspapers & Media	East Jaintia Hills	2023-06-10	3	0
6	Newspapers & Media/2023/Meghalaya/18	Newspapers & Media	East Khasi Hills	2023-06-11	2	0
7	Newspapers & Media/2023/Meghalaya/16	Newspapers & Media	East Jaintia Hills	2023-06-14	3	0
8	Newspapers & Media/2023/Meghalaya/20	Newspapers & Media	East Khasi Hills	2023-06-14	3	0
9	Newspapers & Media/2023/Meghalaya/19	Newspapers & Media	East Jaintia Hills	2023-06-15	3	0
10	Newspapers & Media/2023/Meghalaya/22	Newspapers & Media	West Khasi Hills	2023-06-17	2	2
11	Newspapers & Media/2023/Meghalaya/28	Newspapers & Media	East Khasi Hills	2023-06-17	2	4
12	Newspapers & Media/2023/Meghalaya/29	Newspapers & Media	West Jaintia Hills	2023-06-17	2	0
13	Newspapers & Media/2023/Meghalaya/30	Newspapers & Media	East Jaintia Hills	2023-06-19	3	0
14	Newspapers & Media/2023/Meghalaya/31	Newspapers & Media	West Garo Hills	2023-06-22	2	0
15	Newspapers & Media/2023/Meghalaya/32	Newspapers & Media	Ri Bhoi	2023-06-22	2	0
16	Newspapers & Media/2023/Meghalaya/37	Newspapers & Media	East Jaintia Hills	2023-06-23	3	0
17	Newspapers & Media/2023/Meghalaya/38	Newspapers & Media	East Jaintia Hills	2023-07-03	3	0
18	Newspapers & Media/2023/Meghalaya/56	Newspapers & Media	East Khasi Hills	2023-08-04	3	0
19	Newspapers & Media/2023/Meghalaya/61	Newspapers & Media	East Jaintia Hills	2023-08-10	3	0
20	Newspapers & Media/2023/Meghalaya/70	Newspapers & Media	East Jaintia Hills	2023-08-26	3	0
21	Newspapers & Media/2023/Meghalaya/78	Newspapers & Media	Ri Bhoi	2023-09-26	2	0
22	Newspapers & Media/2023/Meghalaya/84	Newspapers & Media	West Jaintia Hills	2023-10-08	2	4



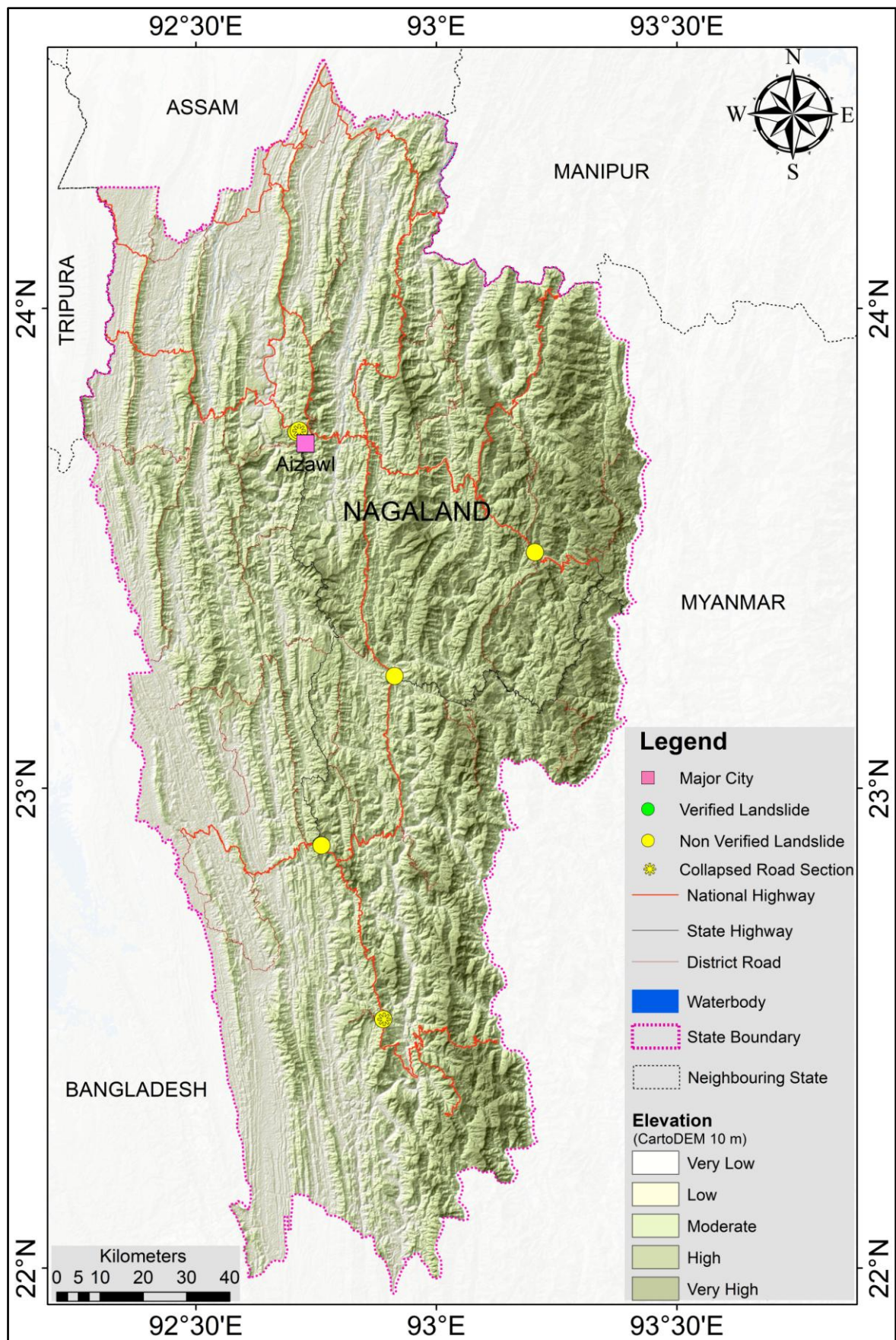
Map 5: Landslide incidents in Meghalaya, 2023

MIZORAM

- 8 landslide incidents recorded in Mizoram. The first landslide incident was recorded on 26th January 2023 and the latest event was recorded on 11th September 2023.
- 4 casualties occurred in Mizoram in 2023.
- NH-6 and NH-54 were among the major road sections affected by landslides in Mizoram in 2023.

Table 9 Summary of landslide incidents in Mizoram

S. No.	Landslide ID	Source	District	Date	Type	Casualty
1	Newspapers & Media/2023/Mizoram/10	Newspapers & Media	Champhai	2023-01-26	2	3
2	Newspapers & Media/2023/Mizoram/24	Newspapers & Media	Aizawl	2023-06-16	2	0
3	Newspapers & Media/2023/Mizoram/41	Newspapers & Media	Aizawl	2023-07-04	2	0
4	Newspapers & Media/2023/Mizoram/55	Newspapers & Media	Aizawl	2023-07-27	2	0
5	Newspapers & Media/2023/Mizoram/57	Newspapers & Media	Lawngtlai	2023-08-08	2	0
6	Newspapers & Media/2023/Mizoram/58	Newspapers & Media	Lunglei	2023-08-08	2	0
7	Newspapers & Media/2023/Mizoram/59	Newspapers & Media	Aizawl	2023-08-09	2	0
8	Newspapers & Media/2023/Mizoram/77	Newspapers & Media	Serchhip	2023-09-11	2	1



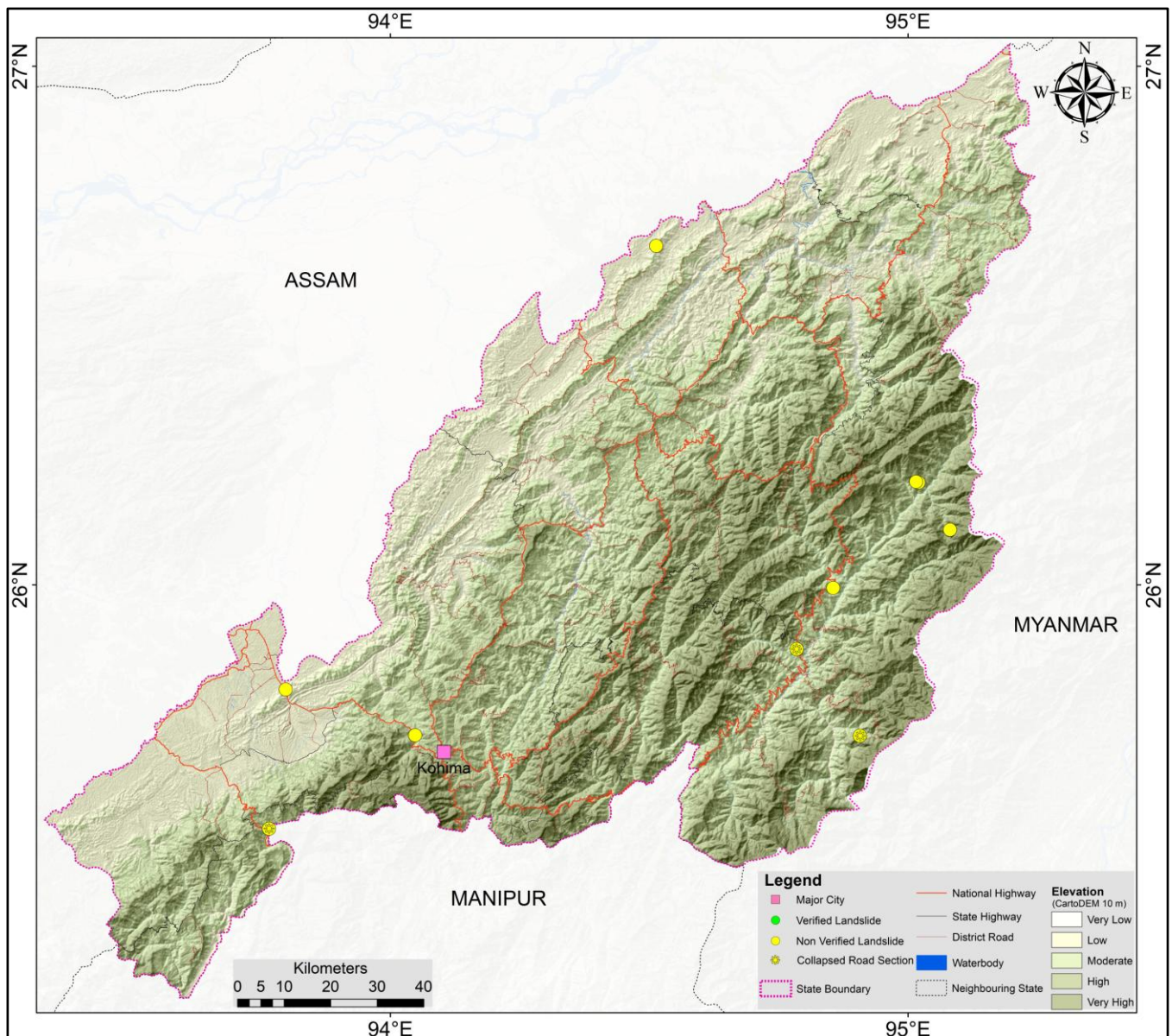
Map 6: Landslide incidents in Mizoram, 2023

NAGALAND

- A total of 10 landslide incidents recorded in Nagaland.
- The first landslide incident for the year was recorded on 4th July 2023 and the last event was recorded on 17th August 2023.
- 3 casualties were reported from Nagaland in 2023.
- NH 29 near Chumoukedima Police Gate was affected by landslides and rock-falls, where 2 casualties and 3 injuries were reported due to the rock fall. Other road sections majorly affected are NH-129A and NH-202.

Table 10 Summary of landslide incidents in Nagaland

S. No.	Landslide ID	Source	District	Date	Type	Casualty
1	Newspapers & Media/2023/Nagaland/39	Newspapers & Media	Chümoukedima	2023-07-04	2	2
2	Newspapers & Media/2023/Nagaland/42	Newspapers & Media	Noklak	2023-07-09	2	0
3	Newspapers & Media/2023/Nagaland/43	Newspapers & Media	Noklak	2023-07-09	2	0
4	Newspapers & Media/2023/Nagaland/44	Newspapers & Media	Noklak	2023-07-09	2	0
5	Newspapers & Media/2023/Nagaland/45	Newspapers & Media	Peren	2023-07-10	2	0
6	Newspapers & Media/2023/Nagaland/51	Newspapers & Media	Kiphire	2023-07-16	2	0
7	Newspapers & Media/2023/Nagaland/52	Newspapers & Media	Kohima	2023-07-17	2	1
8	Newspapers & Media/2023/Nagaland/60	Newspapers & Media	Kiphire	2023-08-09	2	0
9	Newspapers & Media/2023/Nagaland/71	Newspapers & Media	Kiphire	2023-08-25	2	0
10	Newspapers & Media/2023/Nagaland/76	Newspapers & Media	Mokokchung	2023-08-27	2	0



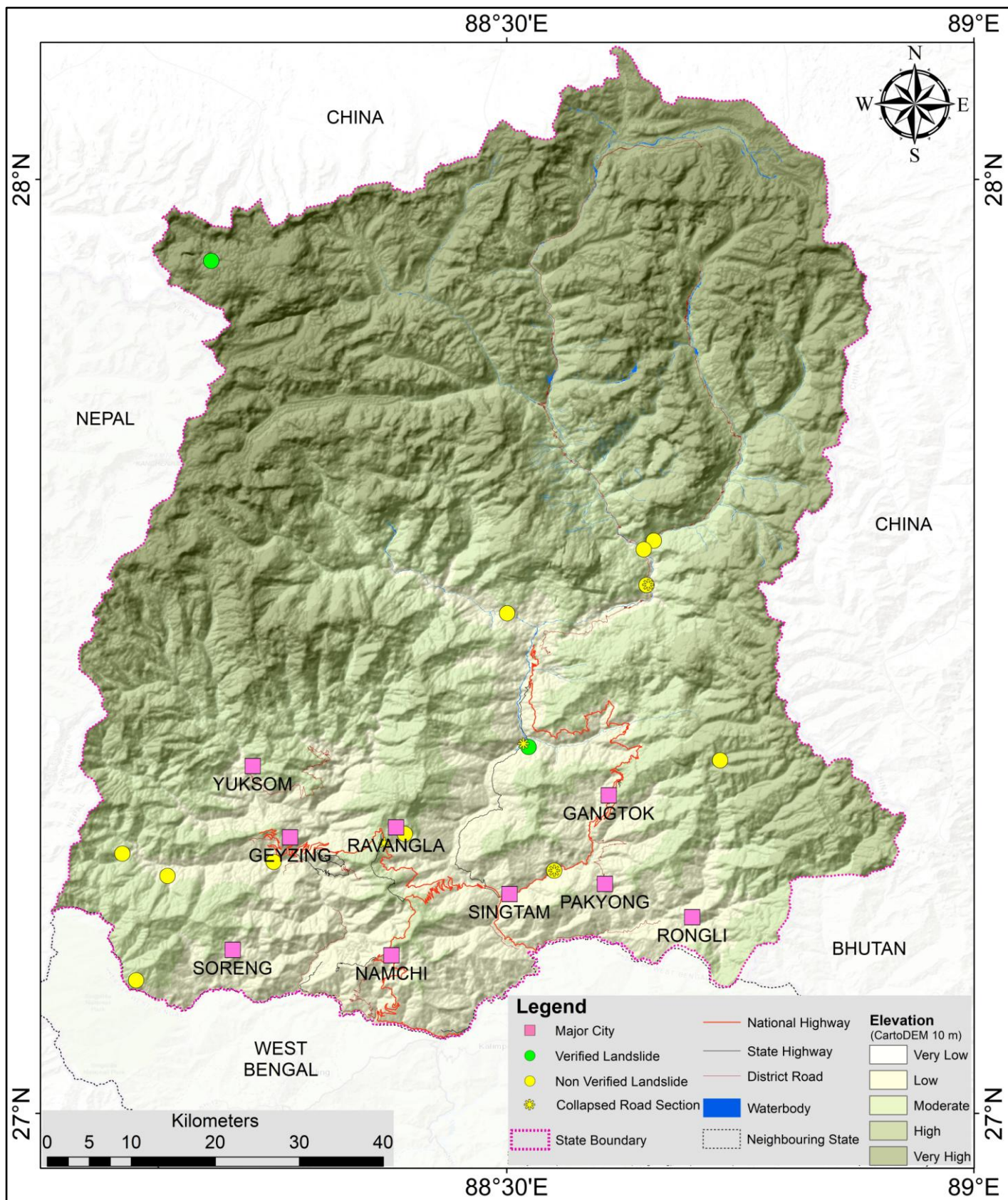
Map 7: Landslide incidents in Nagaland, 2023

SIKKIM

- 14 landslide incidents recorded in Sikkim. The first landslide incident for the year was recorded on 26th March 2023 and the latest event was recorded on 4th October 2023.
- 2 casualties were reported from Sikkim in 2023.
- NH-10, NH-310 and NH-310A were among the major road sections affected by landslide. In NH-10 around 3500 tourists got stranded due to the road blockade caused by landslides.
- South Lhonak lake outburst event and the slope failure has resulted in a huge disaster in the entire Teesta flood plain area taking away more than 40 lives and numerous injuries.

Table 11 Summary of landslide incidents in Sikkim

S. No.	Landslide ID	Source	District	Date	Type	Casualty
1	Newspapers & Media/2023/Sikkim/2	Newspapers & Media	Gangtok	2023-03-26	3	0
2	Newspapers & Media/2023/Sikkim/11	Newspapers & Media	North Sikkim	2023-05-19	2	0
3	Newspapers & Media/2023/Sikkim/21	Newspapers & Media	North Sikkim	2023-06-16	2	0
4	Newspapers & Media/2023/Sikkim/25	Newspapers & Media	West Sikkim	2023-06-18	2	0
5	Newspapers & Media/2023/Sikkim/26	Newspapers & Media	West Sikkim	2023-06-18	2	0
6	Newspapers & Media/2023/Sikkim/27	Newspapers & Media	West Sikkim	2023-06-18	2	0
7	Newspapers & Media/2023/Sikkim/46	Newspapers & Media	North Sikkim	2023-07-12	2	1
8	Newspapers & Media/2023/Sikkim/47	Newspapers & Media	South Sikkim	2023-07-14	2	0
9	Newspapers & Media/2023/Sikkim/48	Newspapers & Media	East Sikkim	2023-07-14	2	0
10	Newspapers & Media/2023/Sikkim/49	Newspapers & Media	North Sikkim	2023-07-14	2	0
11	Newspapers & Media/2023/Sikkim/67	Newspapers & Media	East Sikkim	2023-08-04	2	0
12	Newspapers & Media/2023/Sikkim/68	Newspapers & Media	East Sikkim	2023-08-24	2	0
13	Newspapers & Media/2023/Sikkim/69	Newspapers & Media	West Sikkim	2023-08-25	2	1
14	Newspapers & Media/2023/Sikkim/79	Newspapers & Media	North Sikkim	2023-10-04	1	44 (due to flash flood resulted from Slope failure/GLOF)



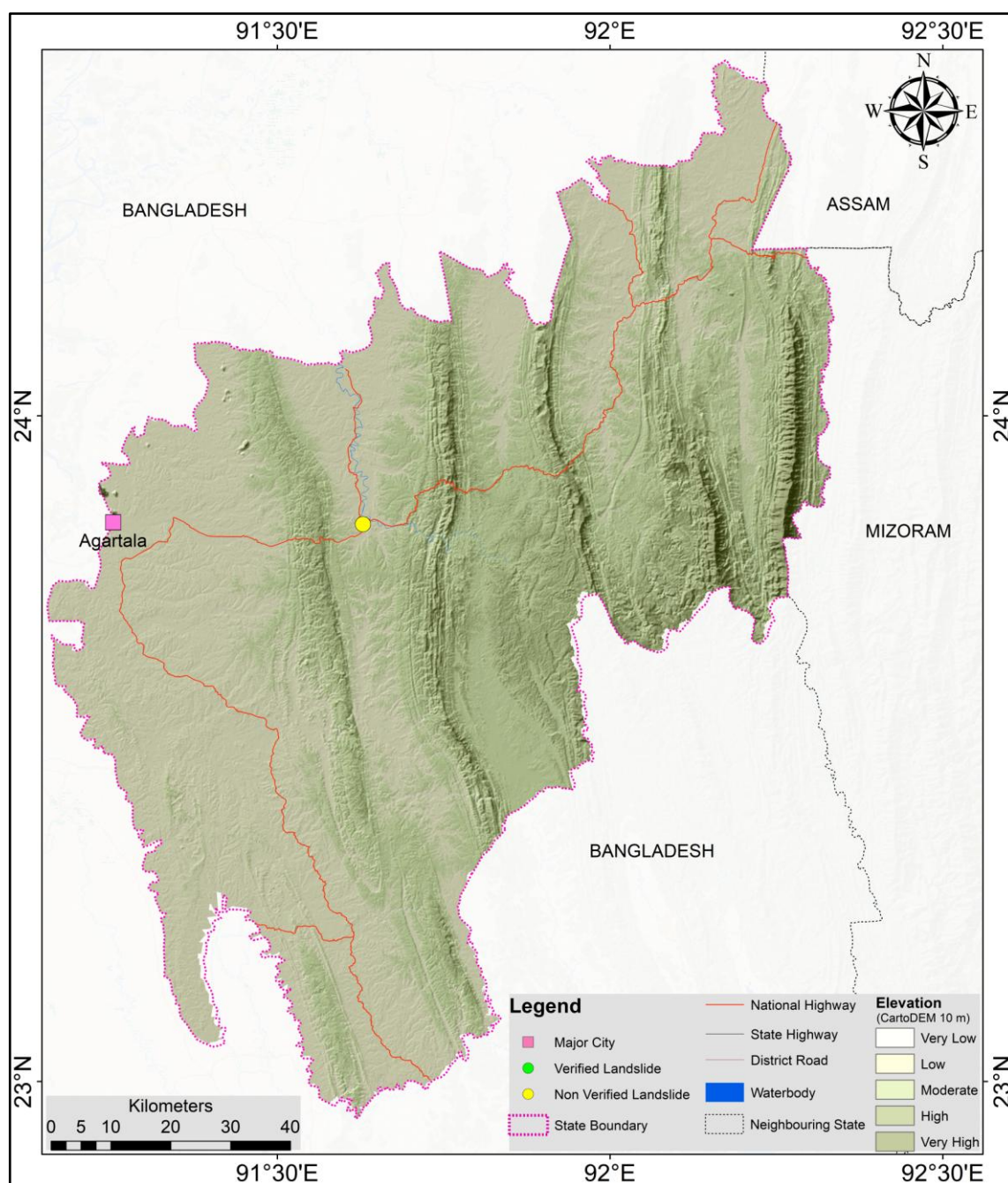
Map 8 Landslide incidents in Sikkim, 2023

TRIPURA

- Only single landslide incident is recorded in Tripura so far which occurred on 31st March 2023.
- No casualties were reported due to the incident. However, 23 tribal families were rendered homeless due to the event.

Table 12 Summary of landslide incidents in Tripura

S. No.	Landslide ID	Source	District	Date	Type	Casualty
1	Newspapers & Media/2023/Tripura/13	Newspapers & Media	Khowai	2023-03-31	2	0



Map 9: Landslide incidents in Tripura, 2023

ANNEXURES

ANNEXURE-I: LANDSLIDE INCIDENT REPORTS

ARUNACHAL PRADESH

Slide ID	Newspapers & Media/2023/ Arunachal Pradesh/1	Slide ID	Newspapers & Media/2023/Arunachal Pradesh/6
Source	Newspapers & Media	Source	Newspapers & Media
Slide No	1	Slide No	6
State	Arunachal Pradesh	State	Arunachal Pradesh
District	Lower Dibang Valley	District	Upper Siang
Name	Roing Anini	Name	Tuting Landslide
Location	Roing Anini Road	Location	Bomdo_Janbo_Road
NH/SH affected	NH 313	NH/SH affected	NH 13
Lat (dd)	28.2333	Lat (dd)	28.78213823
Lon (dd)	95.7776	Lon (dd)	94.86047758
Date of Event	2023-02-24	Date of Event	2023-03-24
Time of Event	--	Time of Event	--
Type of Event	2	Type of Event	3
Casualty	0	Casualty	0
Rainfall (mm)	21.31	Rainfall (mm)	4.30
Slide ID	Newspapers & Media/2023/Arunachal Pradesh/7	Slide ID	Newspapers & Media/2023/Arunachal Pradesh/4
Source	Newspapers & Media	Source	Newspapers & Media
SlideNo	7	SlideNo	4
State	Arunachal Pradesh	State	Arunachal Pradesh
District	Upper Siang	District	Kurung Kumey
Name	Mossing Landslide	Name	Kurung 1
Location	Yingkiong_Tuting_Road	Location	near Kurung bridge
NH/SH affected	Yingkiong_Tuting_Road	NH/SH affected	Koloriang to Parsi Parlo road
Lat (dd)	28.79895656	Lat (dd)	27.89451833
Lon (dd)	94.77570872	Lon (dd)	93.43502779
Date of Event	2023-05-05	Date of Event	2023-05-06
Time of Event	2:00 AM	Time of Event	--
Type of Event	3	Type of Event	3
Casualty	4	Casualty	0
Rainfall (mm)	4.29	Rainfall (mm)	7.562835236

Slide ID	Newspapers & Media/2023/Arunachal Pradesh/5	Slide ID	Newspapers & Media/2023/Arunachal Pradesh/17
Source	Newspapers & Media	Source	Newspapers & Media
SlideNo	5	SlideNo	17
State	Arunachal Pradesh	State	Arunachal Pradesh
District	Kurung Kumey	District	West Kameng
Name	Kurung 2	Name	Tippi Landslide
Location	Near Kurung bridge	Location	Balipara-Chariduar-Tawang Road (Tippi Village)
NH/SH affected	Koloriang to Parsi Parlo road	NH/SH affected	NH 13
Lat (dd)	27.8964705	Lat (dd)	27.026924
Lon (dd)	93.42410335	Lon (dd)	92.610166
Date of Event	2023-05-06	Date of Event	2023-06-12
Time of Event	--	Time of Event	--
Type of Event	3	Type of Event	2
Casualty	0	Casualty	0
Rainfall (mm)	7.56	Rainfall (mm)	97.92
Slide ID	Newspapers & Media/2023/Arunachal Pradesh/33	Slide ID	Newspapers & Media/2023/Arunachal Pradesh/34
Source	Newspapers & Media	Source	Newspapers & Media
SlideNo	33	SlideNo	34
State	Arunachal Pradesh	State	Arunachal Pradesh
District	Papum pare	District	Papum pare
Name	Karsingsa Landslide	Name	Puroik Landslide
Location	NH-415, between Bandardeva and Nirjuli	Location	Puroik Colony, Papu Nalah
NH/SH affected	NH-415	NH/SH affected	Nh-415
Lat (dd)	27.095932	Lat (dd)	27.101535
Lon (dd)	93.828636	Lon (dd)	93.671454
Date of Event	2023-06-23	Date of Event	2023-06-23
Time of Event	--	Time of Event	--
Type of Event	2	Type of Event	2
Casualty	0	Casualty	0
Rainfall (mm)	2.254254371	Rainfall (mm)	3.966167386
Slide ID	Newspapers & Media/2023/Arunachal Pradesh/35	Slide ID	Newspapers & Media/2023/Arunachal Pradesh/36
Source	Newspapers & Media	Source	Newspapers & Media
SlideNo	35	SlideNo	36
State	Arunachal Pradesh	State	Arunachal Pradesh
District	West Kameng	District	West Siang
Name	Sessa Landslide	Name	Pangin Landslide
Location	Balipara-Charduar-Tawang road	Location	Aalo Pangin Landslide
NH/SH affected	--	NH/SH affected	--
Lat (dd)	27.10699	Lat (dd)	28.233721
Lon (dd)	92.525162	Lon (dd)	94.930641
Date of Event	2023-06-23	Date of Event	2023-06-23
Time of Event	--	Time of Event	--
Type of Event	2	Type of Event	2
Casualty	0	Casualty	0
Rainfall (mm)	1.970178811	Rainfall (mm)	0

Slide ID	Newspapers & Media/2023/Arunachal Pradesh/50	Slide ID	Newspapers & Media/2023/Arunachal Pradesh/53
Source	Newspapers & Media	Source	Newspapers & Media
SlideNo	50	SlideNo	53
State	Arunachal Pradesh	State	Arunachal Pradesh
District	Siang district	District	Upper Subansiri
Name	Pangin Landslide	Name	Taliha Landslide
Location	Pangin Pasighat road section	Location	Donyi Colony in Taliha town
NH/SH affected	NH 13	NH/SH affected	--
Lat (dd)	28.095267	Lat (dd)	28.235584
Lon (dd)	95.282822	Lon (dd)	94.144356
Date of Event	2023-07-16	Date of Event	2023-07-28
Time of Event	Morning	Time of Event	--
Type of Event	2	Type of Event	2
Casualty	0	Casualty	0
Rainfall (mm)	78.259454	Rainfall (mm)	76.79615263
Slide ID	Newspapers & Media/2023/Arunachal Pradesh/54	Slide ID	Newspapers & Media/2023/Arunachal Pradesh/64
Source	Newspapers & Media	Source	Newspapers & Media
SlideNo	54	SlideNo	64
State	Arunachal Pradesh	State	Arunachal Pradesh
District	Siang district	District	Lower Siang
Name	NH 13, Pangin Landslide	Name	Siji Landslide
Location	NH 13	Location	Likabali Aalo Road
NH/SH affected	NH 13	NH/SH affected	--
Lat (dd)	28.14251	Lat (dd)	27.717104
Lon (dd)	95.269295	Lon (dd)	94.697075
Date of Event	2023-07-28	Date of Event	2023-08-18
Time of Event	8:30 AM	Time of Event	--
Type of Event	2	Type of Event	2
Casualty	0	Casualty	0
Rainfall (mm)	195.0684341	Rainfall (mm)	4.32477905
Slide ID	Newspapers & Media/2023/Arunachal Pradesh/72	Slide ID	Newspapers & Media/2023/Arunachal Pradesh/73
Source	Newspapers & Media	Source	Newspapers & Media
SlideNo	72	SlideNo	73
State	Arunachal Pradesh	State	Arunachal Pradesh
District	West Kameng	District	West Kameng
Name	Nag Mandir Landslide	Name	Rupa Landslide
Location	Nag Mandir - Rupa Road	Location	Shergaon - Rupa Road
NH/SH affected	NH 13	NH/SH affected	--
Lat (dd)	27.204944	Lat (dd)	27.203191
Lon (dd)	92.547048	Lon (dd)	92.38797
Date of Event	2023-08-27	Date of Event	2023-08-27
Time of Event	--	Time of Event	--
Type of Event	2	Type of Event	2
Casualty	0	Casualty	0
Rainfall (mm)	13.39995506	Rainfall (mm)	18.61701869
Slide ID	Newspapers & Media/2023/Arunachal Pradesh/74	Slide ID	Newspapers & Media/2023/Arunachal Pradesh/75

Source	Newspapers & Media	Source	Newspapers & Media
SlideNo	74	SlideNo	75
State	Arunachal Pradesh	State	Arunachal Pradesh
District	West Kameng	District	West Kameng
Name	Dengzi Landslide	Name	Tenzin Landslide
Location	Dengzi Labor Camp Area	Location	Kalaktang - Tenzin Gaon Road
NH/SH affected	--	NH/SH affected	--
Lat (dd)	27.057521	Lat (dd)	27.09638
Lon (dd)	92.136539	Lon (dd)	92.218027
Date of Event	2023-08-27	Date of Event	2023-08-27
Time of Event	--	Time of Event	--
Type of Event	2	Type of Event	2
Casualty	0	Casualty	0
Rainfall (mm)	203.8414724	Rainfall (mm)	208.6173178
Slide ID	Newspapers & Media/2023/Arunachal Pradesh/85		
Source	Newspapers & Media		
SlideNo	85		
State	Arunachal Pradesh		
District	Lower Subansiri		
Name	Subansiri Landslide		
Location	Near Subansiri Hydropower project dam		
NH/SH affected	--		
Lat (dd)	27.552115		
Lon (dd)	94.258982		
Date of Event	2023-10-27		
Time of Event	11:30 AM		
Type of Event	2		
Casualty	0		
Rainfall (mm)	0		

ASSAM

Slide ID	Newspapers & Media/2023/Assam/8	Slide ID	Newspapers & Media/2023/Assam/23
Source	Newspapers & Media	Source	Newspapers & Media
SlideNo	8	SlideNo	23
State	Assam	State	Assam
District	Dhemaji	District	Kamrup Metro
Name	Subansiri Lower Hydroelectric Project	Name	Dhirenpara Landslide
Location	Gerukamukh	Location	Katahbari Main Road area
NH/SH affected	--	NH/SH affected	--
Lat (dd)	27.55468158	Lat (dd)	26.123803
Lon (dd)	94.25992713	Lon (dd)	91.719487
Date of Event	2023-04-03	Date of Event	2023-06-17
Time of Event	--	Time of Event	--
Type of Event	1	Type of Event	2
Casualty	0	Casualty	1
Rainfall (mm)	9.30	Rainfall (mm)	176.45
Slide ID	Newspapers & Media/2023/Assam/40	Slide ID	Newspapers & Media/2023/Assam/66

Source	Newspapers & Media	Source	Newspapers & Media
SlideNo	40	SlideNo	66
State	Assam	State	Assam
District	Kamrup	District	Cachar
Name	Joramuhuria-Khokhapara road	Name	Lalpani Landslide
Location	Joramuhuria-Khokhapara road	Location	NH-37, Lalpani area
NH/SH affected	--	NH/SH affected	NH-37
Lat (dd)	25.894901	Lat (dd)	24.795353
Lon (dd)	91.43331	Lon (dd)	93.038339
Date of Event	2023-06-29	Date of Event	2023-08-21
Time of Event	--	Time of Event	--
Type of Event	2	Type of Event	2
Casualty	0	Casualty	0
Rainfall (mm)	74.82	Rainfall (mm)	1.96
Slide ID	Newspapers & Media/2023/Assam/80	Slide ID	Newspapers & Media/2023/Assam/81
Source	Newspapers & Media	Source	Newspapers & Media
SlideNo	80	SlideNo	81
State	Assam	State	Assam
District	Kamrup Metro	District	Kamrup Metro
Name	Kharguli Landslide	Name	Bamunimaidam Landslide
Location	Kharguli hill	Location	Bamunimaidam area
NH/SH affected	--	NH/SH affected	--
Lat (dd)	26.19676	Lat (dd)	26.187105
Lon (dd)	91.76544	Lon (dd)	91.793724
Date of Event	06-10-2023	Date of Event	06-10-2023
Time of Event	--	Time of Event	--
Type of Event	2	Type of Event	2
Casualty	0	Casualty	0
Rainfall (mm)	64.05	Rainfall (mm)	64.05

Slide ID	Newspapers & Media/2023/Assam/82	Slide ID	Newspapers & Media/2023/Assam/83
Source	Newspapers & Media	Source	Newspapers & Media
Slide No	82	Slide No	83
State	Assam	State	Assam
District	Kamrup Metro	District	Kamrup Metro
Name	Lechubagan Landslide	Name	Kahilipara Landslide
Location	Hengrabari Area, Guwahati	Location	Kahilipara, Guwahati
NH/SH affected	--	NH/SH affected	--
Lat (dd)	26.152334	Lat (dd)	26.141897
Lon (dd)	91.798065	Lon (dd)	91.75701
Date of Event	07-10-2023	Date of Event	07-10-2023
Time of Event	--	Time of Event	--
Type of Event	2	Type of Event	2
Casualty	0	Casualty	0
Rainfall (mm)	3.36	Rainfall (mm)	3.36

MANIPUR

Slide ID	Newspapers & Media/2023/Manipur/62	Slide ID	Newspapers & Media/2023/Manipur/63
Source	Newspapers & Media	Source	Newspapers & Media
SlideNo	62	SlideNo	63
State	Manipur	State	Manipur
District	Noney	District	Noney
Name	Awangkhum Landslide	Name	Khongsang Landslide
Location	Jiribam Imphal Road	Location	Imphal Silchar Road
NH/SH affected	NH 37	NH/SH affected	NH 37
Lat (dd)	24.820487	Lat (dd)	24.826582
Lon (dd)	93.550414	Lon (dd)	93.477731
Date of Event	2023-08-16	Date of Event	2023-08-16
Time of Event	4:00 PM	Time of Event	--
TypeofEvent	2	Type of Event	2
Casualty	0	Casualty	0
Rainfall (mm)	5.34347082	Rainfall (mm)	4.925795606

Slide ID	Newspapers & Media/2023/Manipur/65
Source	Newspapers & Media
SlideNo	65
State	Manipur
District	Noney
Name	Awangkhum Landslide
Location	Jiribam Imphal Road
NH/SH affected	NH 37
Lat (dd)	24.820481
Lon (dd)	93.55041
Date of Event	2023-08-20
Time of Event	7:00 AM
Type of Event	2
Casualty	0
Rainfall (mm)	20.85353676

MEGHALAYA

Slide ID	Newspapers & Media/2023/Meghalaya/3	Slide ID	Newspapers & Media/2023/Meghalaya/9
Source	Newspapers & Media	Source	Newspapers & Media
SlideNo	3	SlideNo	9
State	Meghalaya	State	Meghalaya
District	East Khasi Hills	District	West Garo Hills
Name	Rngain Landslide	Name	Tura Civil Hospital
Location	Rngain Village, Pynursla	Location	Tura Civil Hospital
NH/SH affected	NH 206	NH/SH affected	--
Lat (dd)	25.371447	Lat (dd)	25.515019

Lon (dd)	91.876558	Lon (dd)	90.202987
Date of Event	2023-04-14	Date of Event	2023-05-06
Time of Event	--	Time of Event	--
Type of Event	3	Type of Event	2
Casualty	2	Casualty	0
Rainfall (mm)	0	Rainfall (mm)	0
Slide ID	Newspapers & Media/2023/Meghalaya/12	Slide ID	Newspapers & Media/2023/Meghalaya/14
Source	Newspapers & Media	Source	Newspapers & Media
SlideNo	12	SlideNo	14
State	Meghalaya	State	Meghalaya
District	East Jaintia Hills	District	East Jaintia Hills
Name	Sonapur Tunnel Landslide	Name	Sonapur Tunnel Landslide
Location	Near Sonapur Tunnel	Location	Near Sonapur Tunnel
NH/SH affected	NH 6	NH/SH affected	NH 6
Lat (dd)	25.105844	Lat (dd)	25.105931
Lon (dd)	92.36018	Lon (dd)	92.360242
Date of Event	2023-05-27	Date of Event	2023-04-05
Time of Event	--	Time of Event	7:00 AM
Type of Event	2	Type of Event	3
Casualty	0	Casualty	0
Rainfall (mm)	0.007794081	Rainfall (mm)	2.958248956
Slide ID	Newspapers & Media/2023/Meghalaya/15	SlideID	Newspapers & Media/2023/Meghalaya/16
Source	Newspapers & Media	Source	Newspapers & Media
SlideNo	15	SlideNo	16
State	Meghalaya	State	Meghalaya
District	East Jaintia Hills	District	East Jaintia Hills
Name	Sonapur Tunnel Landslide	Name	Sonapur Tunnel Landslide
Location	near Sonapur Tunnel	Location	near Sonapur Tunnel
NH/SHaffected	NH 6	NH/SHaffected	NH 6
Lat(dd)	25.105937	Lat(dd)	25.10594
Lon(dd)	92.360245	Lon(dd)	92.360255
DateofEvent	2023-06-10	DateofEvent	2023-06-14
TimeofEvent	--	TimeofEvent	Morning
TypeofEvent	3	TypeofEvent	3
Casualty	0	Casualty	0
Rainfall(mm)	87.10559537	Rainfall(mm)	289.2794716
Slide ID	Newspapers & Media/2023/Meghalaya/18	Slide ID	Newspapers & Media/2023/Meghalaya/19
Source	Newspapers & Media	Source	Newspapers & Media
SlideNo	18	SlideNo	19
State	Meghalaya	State	Meghalaya
District	East Khasi Hills	District	East Jaintia Hills
Name	Pynursla Landslide	Name	Sonapur Tunnel Landslide
Location	Riwar Mihngi circle	Location	near Sonapur Tunnel
NH/SH affected	--	NH/SH affected	NH 6
Lat (dd)	25.309601	Lat (dd)	25.105945
Lon (dd)	91.902608	Lon (dd)	92.360259
Date of Event	2023-06-11	Date of Event	2023-06-15
Time of Event	--	Time of Event	Morning
Type of Event	2	Type of Event	3

Casualty	0	Casualty	0
Rainfall (mm)	7.087581009	Rainfall (mm)	5.790063789
Slide ID	Newspapers & Media/2023/Meghalaya/20	Slide ID	Newspapers & Media/2023/Meghalaya/22
Source	Newspapers & Media	Source	Newspapers & Media
Slide No	20	SlideNo	22
State	Meghalaya	State	Meghalaya
District	East Khasi Hills	District	West Khasi Hills
Name	Rngain Landslide	Name	Dommawlein landslide
Location	Rngain Village, Pynursla	Location	Mawiong-Pyndengrei locality of Nongstoin
NH/SH affected	NH 206	NH/SH affected	--
Lat (dd)	25.371452	Lat (dd)	25.275153
Lon (dd)	91.876551	Lon (dd)	91.366042
Date of Event	2023-06-14	Date of Event	2023-06-17
Time of Event	5:00 AM	Time of Event	4:30 A.M.
Type of Event	3	Type of Event	2
Casualty	0	Casualty	2
Rainfall (mm)	161.3283562	Rainfall (mm)	315.6915713
Slide ID	Newspapers & Media/2023/Meghalaya/28	Slide ID	Newspapers & Media/2023/Meghalaya/29
Source	Newspapers & Media	Source	Newspapers & Media
SlideNo	28	SlideNo	29
State	Meghalaya	State	Meghalaya
District	East Khasi Hills	District	West Jaintia Hills
Name	Laitlarem Landslide	Name	Darrang Landslide
Location	Laitlarem Village area	Location	Darrang Village area
NH/SH affected	--	NH/SH affected	--
Lat (dd)	25.446165	Lat (dd)	25.205927
Lon (dd)	91.57097	Lon (dd)	92.020892
Date of Event	2023-06-17	Date of Event	2023-06-17
Time of Event	--	Time of Event	--
Type of Event	2	Type of Event	2
Casualty	4	Casualty	0
Rainfall (mm)	413.2279502	Rainfall (mm)	237.398367

Slide ID	Newspapers & Media/2023/Meghalaya/30	Slide ID	Newspapers & Media/2023/Meghalaya/31
Source	Newspapers & Media	Source	Newspapers & Media
Slide No	30	Slide No	31
State	Meghalaya	State	Meghalaya
District	East Jaintia Hills	District	West Garo Hills
Name	Sonapur Tunnel Landslide	Name	PA Sangma Stadium Landslide
Location	near Sonapur Tunnel	Location	PA Sangma Stadium
NH/SH affected	NH 6	NH/SH affected	--
Lat (dd)	25.10595	Lat (dd)	25.521879
Lon (dd)	92.3603	Lon (dd)	90.181776
Date of Event	2023-06-19	Date of Event	2023-06-22
Time of Event	--	Time of Event	Morning
Type of Event	3	Type of Event	2
Casualty	0	Casualty	0
Rainfall (mm)	138.4358905	Rainfall (mm)	223.4947659
Slide ID	Newspapers &	Slide ID	Newspapers & Media/2023/Meghalaya/37

Media/2023/Meghalaya/32			
Source	Newspapers & Media	Source	Newspapers & Media
SlideNo	32	Slide No	37
State	Meghalaya	State	Meghalaya
District	Ri Bhoi	District	East Jaintia Hills
Name	Umiam View point Landslide	Name	Sonapur Tunnel Landslide
Location	Umiam view point	Location	near Sonapur Tunnel
NH/SH affected	--	NH/SH affected	NH 6
Lat (dd)	25.687251	Lat (dd)	25.105701
Lon (dd)	91.948666	Lon (dd)	92.361278
Date of Event	2023-06-22	Date of Event	2023-06-23
Time of Event	--	Time of Event	--
Type of Event	2	Type of Event	3
Casualty	0	Casualty	0
Rainfall (mm)	157.0451147	Rainfall (mm)	8.487524291
Slide ID	Newspapers & Media/2023/Meghalaya/38	Slide ID	Newspapers & Media/2023/Meghalaya/56
Source	Newspapers & Media	Source	Newspapers & Media
Slide No	38	Slide No	56
State	Meghalaya	State	Meghalaya
District	East Jaintia Hills	District	East Khasi Hills
Name	Sonapur Tunnel Landslide	Name	Sonapur Tunnel Landslide
Location	near Sonapur Tunnel	Location	Near Sonapur Tunnel
NH/SH affected	NH 6	NH/SH affected	NH 6
Lat (dd)	25.106073	Lat (dd)	25.106083
Lon (dd)	92.36114	Lon (dd)	92.360407
Date of Event	2023-07-03	Date of Event	2023-08-04
Time of Event	Morning	Time of Event	10:00 PM
Type of Event	3	Type of Event	3
Casualty	0	Casualty	0
Rainfall (mm)	69.53567724	Rainfall (mm)	24.50020308
Slide ID	Newspapers & Media/2023/Meghalaya/61	Slide ID	Newspapers & Media/2023/Meghalaya/70
Source	Newspapers & Media	Source	Newspapers & Media
SlideNo	61	Slide No	70
State	Meghalaya	State	Meghalaya
District	East Jaintia Hills	District	East Jaintia Hills
Name	Sonapur Tunnel Landslide	Name	Sonapur Tunnel Landslide
Location	Near Sonapur Tunnel	Location	Near Sonapur Tunnel
NH/SH affected	NH 6	NH/SH affected	NH 6
Lat (dd)	25.106699	Lat (dd)	25.105957
Lon (dd)	92.36123	Lon (dd)	92.360287
Date of Event	2023-08-10	Date of Event	2023-08-26
Time of Event	Night	Time of Event	--
Type of Event	3	Type of Event	3
Casualty	0	Casualty	0
Rainfall (mm)	0.909345005	Rainfall (mm)	46.21286365
Slide ID	Newspapers & Media/2023/Meghalaya/78	Slide ID	Newspapers & Media/2023/Meghalaya/84

Source	Newspapers & Media	Source	Newspapers & Media
Slide No	78	Slide No	84
State	Meghalaya	State	Meghalaya
District	Ri Bhoi	District	West Jaintia Hills
Name	Patharkhmah Landslide	Name	Pynthorlangtein Landslide
Location	Near Patharkhmah Market	Location	Pynthorlangtein Village
NH/SH affected	--	NH/SH affected	--
Lat (dd)	25.88385	Lat (dd)	25.390614
Lon (dd)	91.615265	Lon (dd)	92.180176
Date of Event	26-09-2023	Date of Event	08-10-2023
Time of Event	--	Time of Event	--
Type of Event	2	Type of Event	2
Casualty	0	Casualty	4
Rainfall (mm)	1.515785843	Rainfall (mm)	74.14568969

MIZORAM

Slide ID	Newspapers & Media/2023/Mizoram/10	Slide ID	Newspapers & Media/2023/Mizoram/24
Source	Newspapers & Media	Source	Newspapers & Media
Slide No	10	Slide No	24
State	Mizoram	State	Mizoram
District	Champhai	District	Aizawl
Name	Hermon Veng	Name	Hunthar Landslide
Location	Near Khawzawl	Location	Western outskirts of Aizawl
NH/SH affected	NH 6	NH/SH affected	NH-54
Lat (dd)	23.491709	Lat (dd)	23.745055
Lon (dd)	93.204533	Lon (dd)	92.714248
Date of Event	2023-01-26	Date of Event	2023-06-16
Time of Event	7.30 PM	Time of Event	--
Type of Event	2	Type of Event	2
Casualty	3	Casualty	0
Rainfall (mm)	0.148348405	Rainfall (mm)	368.9439825
Slide ID	Newspapers & Media/2023/Mizoram/41	Slide ID	Newspapers & Media/2023/Mizoram/55
Source	Newspapers & Media	Source	Newspapers & Media
SlideNo	41	Slide No	55
State	Mizoram	State	Mizoram
District	Aizawl	District	Aizawl
Name	Hunthar Landslide	Name	Vaivakawn Landslide
Location	Hunthar in western part of Aizawl	Location	Vaivakawn Zohnuai area
NH/SH affected	NH 54	NH/SH affected	--
Lat (dd)	23.74506	Lat (dd)	23.742421
Lon (dd)	92.71425	Lon (dd)	92.708782
Date of Event	2023-07-04	Date of Event	2023-07-27
Time of Event	--	Time of Event	--

Type of Event	2	Type of Event	2
Casualty	0	Casualty	0
Rainfall (mm)	0.956700019	Rainfall (mm)	43.21284914
Slide ID	Newspapers & Media/2023/Mizoram/57	Slide ID	Newspapers & Media/2023/Mizoram/58
Source	Newspapers & Media	Source	Newspapers & Media
Slide No	57	Slide No	58
State	Mizoram	State	Mizoram
District	Lawngtlai	District	Lunglei
Name	Lawngtlai Landslide	Name	Lunglei Landslide
Location	Near Lawngtlai town	Location	Near Lunglei town
NH/SH affected	--	NH/SH affected	--
Lat (dd)	22.518533	Lat (dd)	22.880829
Lon (dd)	92.889308	Lon (dd)	92.760969
Date of Event	2023-08-08	Date of Event	2023-08-08
Time of Event	--	Time of Event	--
Type of Event	2	Type of Event	2
Casualty	0	Casualty	0
Rainfall (mm)	95.14827251	Rainfall (mm)	71.49705807
Slide ID	Newspapers & Media/2023/Mizoram/59	Slide ID	Newspapers & Media/2023/Mizoram/77
Source	Newspapers & Media	Source	Newspapers & Media
Slide No	59	Slide No	77
State	Mizoram	State	Mizoram
District	Aizawl	District	Serchhip
Name	Hunthar Landslide	Name	Keitum Landslide
Location	Western outskirts of Aizawl	Location	Keitum Village
NH/SH affected	NH 6	NH/SH affected	--
Lat (dd)	23.745833	Lat (dd)	23.233976
Lon (dd)	92.713846	Lon (dd)	92.91265
Date of Event	2023-08-09	Date of Event	11-09-2023
Time of Event	3:00 AM	Time of Event	--
Type of Event	2	Type of Event	2
Casualty	0	Casualty	1
Rainfall (mm)	33.47074337	Rainfall (mm)	32.64047092

NAGALAND

Slide ID	Newspapers & Media/2023/Nagaland/39	Slide ID	Newspapers & Media/2023/Nagaland/42
Source	Newspapers & Media	Source	Newspapers & Media
Slide No	39	Slide No	42
State	Nagaland	State	Nagaland
District	Chümoukedima	District	Noklak
Name	Pakala Pahar landslide	Name	Choklangan Landslide
Location	near Chümoukedima police post	Location	Choklangan village in Noklak district
NH/SH affected	NH 29	NH/SH affected	--
Lat (dd)	25.798326	Lat (dd)	26.106224

Lon (dd)	93.797729	Lon (dd)	95.080945
Date of Event	2023-07-04	Date of Event	2023-07-09
Time of Event	5:00 PM	Time of Event	1:00 AM
Type of Event	2	Type of Event	2
Casualty	2	Casualty	0
Rainfall (mm)	84.62119811	Rainfall (mm)	56.87882877
Slide ID	Newspapers & Media/2023/Nagaland/43	Slide ID	Newspapers & Media/2023/Nagaland/44
Source	Newspapers & Media	Source	Newspapers & Media
SlideNo	43	Slide No	44
State	Nagaland	State	Nagaland
District	Noklak	District	Noklak
Name	Wonthai Landslide	Name	Ehoukeh King
Location	Wonthai area	Location	Ehoukeh King, Choklangan
NH/SH affected	--	NH/SH affected	--
Lat (dd)	26.197297	Lat (dd)	26.199008
Lon (dd)	95.020282	Lon (dd)	95.015413
Date of Event	2023-07-09	Date of Event	2023-07-09
Time of Event	--	Time of Event	--
Type of Event	2	Type of Event	2
Casualty	0	Casualty	0
Rainfall (mm)	56.55963898	Rainfall (mm)	56.55963898
Slide ID	Newspapers & Media/2023/Nagaland/45	Slide ID	Newspapers & Media/2023/Nagaland/51
Source	Newspapers & Media	Source	Newspapers & Media
SlideNo	45	Slide No	51
State	Nagaland	State	Nagaland
District	Peren	District	Kiphire
Name	Iron bridge landslide	Name	Kiphire Landslide
Location	Peren-Maram road	Location	Shamator Kiphire Road Section
NH/SH affected	NH-129A	NH/SH affected	NH 202
Lat (dd)	25.529813	Lat (dd)	25.994082
Lon (dd)	93.76566	Lon (dd)	94.85535
Date of Event	2023-07-10	Date of Event	2023-07-16
Time of Event	Morning	Time of Event	--
Type of Event	2	Type of Event	2
Casualty	0	Casualty	0
Rainfall (mm)	4.89017205	Rainfall (mm)	175.991999
Slide ID	Newspapers & Media/2023/Nagaland/52	Slide ID	Newspapers & Media/2023/Nagaland/60
Source	Newspapers & Media	Source	Newspapers & Media
Slide No	52	Slide No	60
State	Nagaland	State	Nagaland
District	Kohima	District	Kiphire
Name	Meriema Landslide	Name	Kiphire Town Landslide
Location	Meriema area	Location	Road leading to Pungro town from Kiphire

NH/SH affected	--	NH/SH affected	NH 202
Lat (dd)	25.709988	Lat (dd)	25.876143
Lon (dd)	94.047805	Lon (dd)	94.784753
Date of Event	2023-07-17	Date of Event	2023-08-09
Time of Event	--	Time of Event	Night
Type of Event	2	Type of Event	2
Casualty	1	Casualty	0
Rainfall (mm)	47.45567152	Rainfall (mm)	146.652082
Slide ID	Newspapers & Media/2023/Nagaland/71	Slide ID	Newspapers & Media/2023/Nagaland/76
Source	Newspapers & Media	Source	Newspapers & Media
Slide No	71	Slide No	76
State	Nagaland	State	Nagaland
District	Kiphire	District	Mokokchung
Name	Mimi Landslide	Name	Noklangsangba Landslide
Location	Indo-Myanmar Border area around Mimi Village	Location	Noklangsangba Government High School area
NH/SH affected	--	NH/SH affected	--
Lat (dd)	25.708869	Lat (dd)	26.653847
Lon (dd)	94.906812	Lon (dd)	94.51346
Date of Event	2023-08-25	Date of Event	27-08-2023
Time of Event	--	Time of Event	--
Type of Event	2	Type of Event	2
Casualty	0	Casualty	0
Rainfall (mm)	7.4942718	Rainfall (mm)	8.057369572

SIKKIM

Slide ID	Newspapers & Media/2023/Sikkim/2	Slide ID	Newspapers & Media/2023/Sikkim/11
Source	Newspapers & Media	Source	Newspapers & Media
SlideNo	2	Slide No	11
State	Sikkim	State	Sikkim
District	Gangtok	District	North Sikkim
Name	Sokpay Landslide	Name	Chunghang Landslide
Location	Sokpay Village	Location	Chunghang_Lachung_Road
NH/SH affected	Dikchu-Rakdong Road	NH/SH affected	Chunghang_Lachung_Road
Lat (dd)	27.39235464	Lat (dd)	27.613235
Lon (dd)	88.52320007	Lon (dd)	88.657305
Date of Event	2023-03-26	Date of Event	2023-05-19
Time of Event	--	Time of Event	--
Type of Event	3	Type of Event	2
Casualty	0	Casualty	0
Rainfall (mm)	0.669916422	Rainfall (mm)	13.09883959
Slide ID	Newspapers & Media/2023/Sikkim/21	Slide ID	Newspapers & Media/2023/Sikkim/25
Source	Newspapers & Media	Source	Newspapers & Media
SlideNo	21	Slide No	25
State	Sikkim	State	Sikkim

District	North Sikkim	District	West Sikkim
Name	Chungthang 2 Landslide	Name	Kalej Khola landslide
Location	Along Chungthnag Road	Location	Lower Sapung area
NH/SH affected	NH 10	NH/SH affected	--
Lat (dd)	27.603889	Lat (dd)	27.26933
Lon (dd)	88.64646	Lon (dd)	88.250366
Date of Event	2023-06-16	Date of Event	2023-06-18
Time of Event	--	Time of Event	--
Type of Event	2	Type of Event	2
Casualty	0	Casualty	0
Rainfall (mm)	3.215050509	Rainfall (mm)	66.43408392
Slide ID	Newspapers & Media/2023/Sikkim/26	Slide ID	Newspapers & Media/2023/Sikkim/27
Source	Newspapers & Media	Source	Newspapers & Media
SlideNo	26	Slide No	27
State	Sikkim	State	Sikkim
District	West Sikkim	District	West Sikkim
Name	Dentam Landslide	Name	Lower Okhrey Landslide
Location	Dentam Sub Division under Gyalshing district	Location	Sombaria area
NH/SH affected	--	NH/SH affected	--
Lat (dd)	27.254354	Lat (dd)	27.142583
Lon (dd)	88.136778	Lon (dd)	88.103311
Date of Event	2023-06-18	Date of Event	2023-06-18
Time of Event	--	Time of Event	--
Type of Event	2	Type of Event	2
Casualty	0	Casualty	0
Rainfall (mm)	47.51506926	Rainfall (mm)	87.17961762

Slide ID	Newspapers & Media/2023/Sikkim/46	Slide ID	Newspapers & Media/2023/Sikkim/47
Source	Newspapers & Media	Source	Newspapers & Media
Slide No	46	Slide No	47
State	Sikkim	State	Sikkim
District	North Sikkim	District	South Sikkim
Name	Lingzya Busty Landslide	Name	Rafung Khola Landslide
Location	Lingzya Busty of Upper Dzongu	Location	Rafung Khola area
NH/SH affected	--	NH/SH affected	--
Lat (dd)	27.535576	Lat (dd)	27.299226
Lon (dd)	88.500328	Lon (dd)	88.390868
Date of Event	2023-07-12	Date of Event	2023-07-14
Time of Event	8:30 AM	Time of Event	--
Type of Event	2	Type of Event	2
Casualty	1	Casualty	0
Rainfall (mm)	5.673429807	Rainfall (mm)	53.43282726
Slide ID	Newspapers & Media/2023/Sikkim/48	Slide ID	Newspapers & Media/2023/Sikkim/49
Source	Newspapers & Media	Source	Newspapers & Media
Slide No	48	Slide No	49
State	Sikkim	State	Sikkim
District	East Sikkim	District	North Sikkim

Name	9th Mile Landslide	Name	Theng Landslide
Location	9th Mile area	Location	Theng Tunnel area
NH/SH affected	NH 310	NH/SH affected	NH 310A
Lat (dd)	27.378152	Lat (dd)	27.565792
Lon (dd)	88.728199	Lon (dd)	88.649425
Date of Event	2023-07-14	Date of Event	2023-07-14
Time of Event	--	Time of Event	--
Type of Event	2	Type of Event	2
Casualty	0	Casualty	0
Rainfall (mm)	14.54079002	Rainfall (mm)	5.525783284
Slide ID	Newspapers & Media/2023/Sikkim/67	Slide ID	Newspapers & Media/2023/Sikkim/68
Source	Newspapers & Media	Source	Newspapers & Media
SlideNo	67	Slide No	68
State	Sikkim	State	Sikkim
District	East Sikkim	District	East Sikkim
Name	32 No. Landslide	Name	32 No. Landslide
Location	National Highway NH-10, connecting Sikkim with the mainland and capital Gangtok	Location	National Highway NH-10, connecting Sikkim with the mainland and capital Gangtok
NH/SH affected	NH 10	NH/SH affected	NH 10
Lat (dd)	27.259994	Lat (dd)	27.259936
Lon (dd)	88.55059	Lon (dd)	88.550684
Date of Event	2023-08-04	Date of Event	2023-08-24
Time of Event	--	Time of Event	--
Type of Event	2	Type of Event	2
Casualty	0	Casualty	0
Rainfall (mm)	1.98	Rainfall (mm)	21.48876357
Slide ID	Newspapers & Media/2023/Sikkim/69	Slide ID	Newspapers & Media/2023/Sikkim/79
Source	Newspapers & Media	Source	Newspapers & Media
Slide No	69	Slide No	79
State	Sikkim	State	Sikkim
District	West Sikkim	District	North Sikkim
Name	Lingzing Landslide	Name	Lhonak Landslide
Location	Maneybong Dentam constituency	Location	Near South Lhonak Lake
NH/SH affected	--	NH/SH affected	--
Lat (dd)	27.278215	Lat (dd)	27.912811
Lon (dd)	88.08848	Lon (dd)	88.18362
Date of Event	2023-08-25	Date of Event	04-10-2023
Time of Event	Morning	Time of Event	--
Type of Event	2	Type of Event	1
Casualty	1	Casualty	44 due to flash flood
Rainfall (mm)	4.418032641	Rainfall (mm)	1.658

TRIPURA

Slide ID	Newspapers & Media/2023/Tripura/13
Source	Newspapers & Media
Slide No	13
State	Tripura
District	Khowai
Name	Teliamura Landslide
Location	45 mile area of Atharamura hill range
NH/SH affected	--
Lat (dd)	23.837263
Lon (dd)	91.62872
Date of Event	2023-03-31
Time of Event	Night
Type of Event	2
Casualty	0
Rainfall (mm)	30.87335595

ANNEXURE-II: NEWSPAPER (CLIPPINGS & DIGITAL ARCHIVE)

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News / India News / Landslide damages part of Subansiri hydroelectric project

Landslide damages part of Subansiri hydroelectric project

By Jayashree Nandi, New Delhi

Oct 28, 2023 06:56 AM IST

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This is the second landslide at the same location. A similar landslide impacted the project in March



The project is scheduled to begin operations from March 2024 (HT Photo)

A landslide has damaged part of the under-construction 2000 MW Lower Subansiri Hydroelectric project in Arunachal Pradesh and Assam which has affected the flow in

NP NagalandPost

Home State Regional National International Infotainment Sport Business

Rain-triggered landslides at Noklak district

July 10, 2023



houses washed away by landslide at Phokkham village

मणिपुर: भूस्खलन से इम्फाल-सिलचर मार्ग अवरुद्ध, 500 से अधिक सामान लदे ट्रक फंसे

अमर उजाला ब्यूरो, इम्फाल Published by: स्वप्निल शशांक Updated Sun, 20 Aug 2023 12:00 PM IST

लेटेस्ट अपडेट्स के लिए फॉलो करें

सार

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सूत्रों के मुताबिक, रविवार सुबह करीब 7 बजे ताजा और भारी बारिश के कारण भूस्खलन हुआ। फिलहाल सड़क साफ करने का काम जारी है।



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Earthquake in Tibet triggers landslide & rockfall in Arunachal

Mukut Das / TNN / Updated: Jul 29, 2023, 07:37 IST

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Earthquake in Tibet triggers landslide & rockfall in Arunachal



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Home / Cities / Others / Meghalaya: 2 siblings, minors, killed in landslide in West ...

Meghalaya: 2 siblings, minors, killed in landslide in West Khasi Hills district

By David Laitphlang

Jun 17, 2023 06:00 PM IST

Several houses and shops located close to Nondein River were also severely affected by the heavy rainfall, said officials, adding that they have cut off road communication in several localities amid flash floods

ANNEXURE-III: ONLINE MEDIA

NORTHEAST LIVE
NORTHEAST INDIA'S ONLY ENGLISH AND HINDI SATELLITE NEWS CHANNEL

Arunachal: Landslides block Balipara-Chariduar-Tawang road. Here are roads to avoid

First Published: 28th August, 2023 21:50 IST

Several patches of Balipara-Chariduar-Tawang road have been severely damaged near Nag Mandir, Birpur, 91 RCC and Rupa Mod

Northeast Live Digital Desk

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News / Sikkim / Sikkim: Massive landslide claims life of 4-year-old in Lower Lungchok

Sikkim: Massive landslide claims life of 4-year-old in Lower Lungchok

A landslide took the life of an infant in Lower Lungchok under the Maneybong- Dentam constituency under West Sikkim.

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Mud And Rock Slide At Meghalaya's Sonapur Tunnel, Hundreds Of Vehicles Left Stranded; Watch Visuals

Visuals showed mud and rocks cascade at the Sonapur Tunnel located on National Highway NH-06 in the East Jaintia Hills of Meghalaya.

FPJ Web Desk | Updated: Saturday, August 26, 2023, 04:29 PM IST



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Nagaland

One killed in rain-induced landslide in Nagaland

The incident happened in the Meriema area of Nagaland when construction labourers were working at the site, an official said

by Press Trust of India
July 18, 2023



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Home / News / India / Landslides Due To Heavy Rain Cut Off Nathula Pass, Other Parts Of Sikkim

Landslides Due To Heavy Rain Cut Off Nathula Pass, Other Parts Of Sikkim

Six minor landslides in Sikkim's Rafung Khola have also cut off tourist destinations Lachung and Lachen from the rest of the state.

By: ABP News Bureau | Updated at : 14 Jul 2023 03:59 PM (IST)



Thursday, June 22, 2023

YOUR VOICE..YOUR STORY

THE MEGHALAYAN

HOME MEGHALAYA NORTHEAST NATIONAL INTERNATIONAL SPORTS FEATURES EDITORIAL

Home > Top Story

TOP STORY

Section of PA Sangma stadium collapses due to incessant rains

A section of the stadium that did not have any retaining wall witnessed a major landslip bringing down tons of soil from the field above.

By The Meghalayan Bureau June 22, 2023

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146,751 Fans

ANNEXURE-IV: INVENTORY ATTRIBUTES

Slide ID	Unique Identifier for Landslide
Source	Newspaper/Media, SDMA, GSI or From Field
Slide No	Serial number
State	State in NER where landslide occurred
District	District where landslide occurred
Name	Name of the event
Location	Specific location from SDMA reports. General location – neighbourhood, village name etc from newspapers
NH/SH Cutoff/Affected	If National Highway, State Highway or any other important road was cutoff or affected
Latitude	Exact Latitude from GSI/SDMA reports General Latitude from Newspaper Articles
Longitude	Exact Longitude from GSI/SDMA reports General Longitude from Newspaper Articles
Date of Event	Date of the event
Time of Event	Time of event wherever possible
Type of Event	Type 1, 2 or 3 (explained in the next section)
Casualty	Casualty figures
Rainfall (24 hrs)	24 hours rainfall prior to the event

ANNEXURE-V: CODE SAMPLE FOR LANDSLIDE DETECTION

```
*C:\Users\NESAC\Desktop\2020\python\gee_code.py - Notepad++
File Edit Search View Encoding Language Settings Tools Macro Run Plugins Window ?

gee_code.py
1 import ee
2 from ee_plugin import Map
3
4 #Code for displaying Pre and Post Imagery
5
6 # Processing polygon_ls
7 polygon_ls = "users/tusharmontfort/senapati"
8
9 aoi = ee.FeatureCollection(polygon_ls);
10
11 # ENTER THE TIME FRAME for the first image
12 start_date1 = '2020-10-01'
13 end_date1 = '2020-10-23'
14
15
16
17 # Import the Sentinel 2 collection as sentinel 2
18 sentinel2_1 = ee.ImageCollection("COPERNICUS/S2")
19
20 # Import the boundary
21 ls_bnd = ee.FeatureCollection(polygon_ls)
22
23 # Filter the image collection using filterBounds() and filterDate() method.
24 # Sort the collection by cloud cover metadata
25 image1 = sentinel2_1 \
26     .filterBounds(ls_bnd) \
27     .filterDate(start_date1, end_date1) \
28     .sort('CLOUDY_PIXEL_PERCENTAGE', False) \
29     .map(lambda image1: image1.addBands(image1.metadata('system:time_start')))\
30     .mosaic()\
31     .clip(ls_bnd)
32
33
34 #TRUE COLOUR COMPOSITE
35 visualization_TCC = {
36     'min': 0.0,
37     'max': 3000,
38     'bands': ['B4', 'B3', 'B2']
39 };
40
41 # FALSE COLOUR COMPOSITE
42 visualization_FCC = {
43     'min': 0.0,
44     'max': 3000,
45     'bands': ['B8', 'B4', 'B3']
46 };
47
48 LS_Scene = ee.Image(image1).select(['B4', 'B3', 'B2', 'B8']);
49
50 Map.addLayer(LS_Scene, visualization_TCC, 'true_color_composite_pre');
51 Map.addLayer(LS_Scene, visualization_FCC, 'False_color_composite_pre');
```

ANNEXURE-VI: CODE SAMPLE FOR LANDSLIDE NEWS

```
File Edit Search View Encoding Language Settings Tools Macro Run Plugins Window ?
Landslide_Inventory_2023.csv Landslide_Inventory_11-09-2023.csv change.log new 1 X

1 import requests
2 from bs4 import BeautifulSoup
3 import pandas as pd
4
5 # Define the search query and URL for Google News
6 query = 'landslide OR landslides OR rockfall OR landslip OR rockslide OR mudslide OR भूस्खलन OR debris slide'
7 url = f"https://news.google.com/search?q={query}&hl=en-IN&gl=IN&ceid=IN&3Aen OR https://twitter.com/search?q={query}&hl=en-IN&gl=IN&ceid=IN&3Aen"
8
9 # Send a GET request to the Google News URL
10 response = requests.get(url)
11
12 # Parse the HTML content using BeautifulSoup
13 soup = BeautifulSoup(response.content, 'html.parser')
14
15 # Find the relevant elements containing the news articles
16 articles = soup.find_all('div', {'class': 'NiLawe'})
17
18 # Create a list to store the captured data
19 data = []
20
21 # Extract the desired information from each article
22 for article in articles:
23     title_element = article.find('h3')
24     source_element = article.find('a', {'class': 'wEwycr AVN2gc uQIVzc Sksgp'})
25     timestamp_element = article.find('time')
26     description_element = article.find('div', {'class': 'DalOTb'})
27     link_element = article.find('a', {'class': 'VDXfz'})
28
29     # Check if the elements exist before accessing their attributes
30     title = title_element.text.strip() if title_element else ''
31     source = source_element.text.strip() if source_element else ''
32     timestamp = timestamp_element['datetime'].strip() if timestamp_element else ''
33     description = description_element.text.strip() if description_element else ''
34     link = f"https://news.google.com{link_element['href']}" if link_element else ''
35
36     data.append({'title': title, 'url': link, 'post_time': timestamp, 'source': source, 'description': description, })
37
38 # Create a Pandas DataFrame to store the captured data
39
40 current_df = pd.DataFrame(data)
41 print(len(current_df))
42 current_df.to_csv('D:/LANDSLIDE_PYTHON/Landslide_news.csv')
```